

THE COPPERBELT UNIVERSITY

“Knowledge and Service”



SCHOOL OF ENGINEERING

THE DEPARTMENT OF MECHANICAL ENGINEERING

DESIGN AND IMPROVEMENT OF A PERSONAL TRANSPORTER

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This project report is submitted in partial fulfilment of the requirements for the award of a Bachelor of Engineering in (Hons) Mechatronics Engineering & Mechanical Engineering

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DECLARATION

We hereby declare that we carried out the work reported in this report in the Department of Mechanical Engineering, Copperbelt University, under the supervision of Mr. George Mugala.

We solemnly declare that to the best of our knowledge, no part of this report has been submitted here or elsewhere in a previous application for award of a degree. All the sources of knowledge used have been duly acknowledged.

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APPROVAL

This is to certify that the project titled **Design and Improvement of a Personal Transporter** carried out by Rene Irasubiza and Naaman S. Mwela has been read and approved for meeting part of the requirements and regulations governing the MC 500 Projects course in Mechatronics Engineering at the Copperbelt University, Kitwe, Zambia.

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ABSTRACT

A personal transporter is a compact, electrically powered vehicle designed for short-distance mobility. It provides an environmentally friendly and convenient alternative to conventional transport, allowing individuals to move independently. While such vehicles are widely used in developed countries, their high-cost limits accessibility in regions like Zambia.

In Zambia, there is a strong need for affordable and reliable transportation, particularly for delivery services and personal mobility. This project aims to address this gap by developing a low-cost, locally made personal transporter that is practical, dependable, and easy to use. By providing an affordable solution, the transporter can reduce transportation challenges and support sustainable mobility in the local context.

The proposed design improves upon a previous student prototype that faced limitations in motor control. This model uses a digital Proportional-Integral-Derivative (PID) control system implemented on an Arduino Uno to achieve smooth acceleration, steady speed, and better stability. Brushless hub motors powered by a 36V lithium rechargeable battery were selected and sized according to expected loads and performance requirements. A hall sensor embedded in the hub wheel provides speed feedback, while a potentiometer allows user input control. The transporter is designed to achieve a speed range of up to 26 km/h and carry a load of approximately 70 kg, combining affordability with functionality and reliability.

The final design demonstrates improved motor control, stability, and usability compared to the earlier prototype. This project presents the design, development, and improvement of a three-wheeled personal transporter intended for both mobility assistance and lightweight delivery applications. The work followed a structured engineering approach, beginning with conceptual design and progressing through simulation, prototyping, and testing. Key areas of focus included mechanical stability, electrical system optimization, and motor control through proportional–integral–derivative (PID) tuning. The final design demonstrates significant improvements in motor control response time (reduced overshoot by approximately 35%), stability under load (able to carry up to 65 kg compared to 55 kg in the initial prototype), and usability, with more intuitive turning and simplified control integration. User feedback validated the improvements, particularly the enhanced steering system and battery monitoring display, which increased confidence and comfort during operation. Overall, the refined prototype shows strong potential for practical implementation as a low-cost mobility solution, while future work should focus on extending battery life, improving load capacity beyond 70 kg, and refining the handlebar for height adjustability.

KEYWORDS: Proportional Integral Derivative (PID), Brushless Hub Motor, Lithium Battery, Hall sensor.

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CHAPTER 1: INTRODUCTION

1.0 BACKGROUND

Three-wheeled personal transporters are compact vehicles designed to offer short-distance mobility with enhanced stability compared to two-wheeled vehicles [1]. They are particularly useful for individuals with mobility impairments, urban commuters, or delivery personnel who require reliable and easy-to-use transport solutions [2]. Over the past few decades, three-wheeled designs have evolved from small, economical vehicles in post-war Europe to modern electric scooters capable of supporting daily urban travel [3], [4].



Figure 1. 1 Shows a delivery man on a 3-wheel Personal Transporter

Studies have shown that three-wheeled personal transporters improve accessibility and independence for users with mobility challenges [5]. Langrish et al. [6] demonstrated that personal transporters such as the Segway can serve as alternative mobility devices for individuals with disabilities, offering greater comfort and reducing physical strain. Likewise, Ando and Li [7] highlighted that the public's acceptance and ergonomic suitability of three-wheeled mobility vehicles are critical factors for widespread adoption, emphasizing the importance of human-centered design.



Figure 1. 2 An elderly disenabled lady using a aided by electric transporter

Modern electric three-wheeled scooters integrate advanced technologies to improve functionality, efficiency, and user experience [8]. For instance, hub motors powered by rechargeable batteries are commonly used to drive the wheels, while electronic control systems regulate speed, torque, and stability [9]. Studies by Magibalan [10] and Faris et al. [11] have explored design optimization, motor sizing, and load capacity to enhance performance and affordability. Sensors such as hall-effect sensors are employed for wheel-speed feedback, enabling precise control of acceleration and braking. These design considerations allow the transporters to carry loads up to 70 kg and achieve speeds suitable for urban environments, while maintaining stability and safety.



Figure 1. 3 A Hub Motor Wheel

The continued development of three-wheeled personal transporters reflects a growing demand for sustainable, low-cost, and user-friendly mobility solutions [12]. By combining ergonomic design, electric propulsion, and digital control systems, these vehicles offer a practical alternative to conventional two- or four-wheeled vehicles, particularly for users requiring additional stability or for delivery applications in congested urban areas [13].

1.1 PROBLEM STATEMENT

In Zambia, there is a clear need for cheap and reliable transportation, especially in areas like delivery services where many people cannot afford expensive personal transporters. Last year, a student group worked on this problem and came up with a Do It Yourself (DIY) personal transporter that was much cheaper compared to what is already on the market. Their design showed that it is possible to build a functional and affordable transporter. However, the model they made had challenges with motor control, which made it difficult to maintain steady speed and smooth performance. Because of this limitation, the transporter was not fully reliable for everyday use. This year, our group is improving on that design by introducing a PID control system. With PID, the transporter will have better speed regulation, smoother acceleration, and more stability under different loads and road conditions. In this way, the project aims to provide not only a low-cost solution but also one that is reliable and practical for the Zambian transport sector.

1.2 PROJECT AIM

The aim of this project is to design and build a low-cost three-wheeled personal transporter that uses a digital Proportional-Integral-Derivative (PID) control system. The goal is to make the transporter more stable, reliable, and affordable so that it can be practical for everyday use in Zambia.

1.2 PROJECT OBJECTIVES

The objectives of this project are;

- To design a three-wheeled transporter using simple and affordable materials that can be sourced locally.
- To implement a Proportional Integral Derivative (PID) digital Controller on Arduino.
- To test the transporter for speed, load, and stability and compare the results with the earlier prototype.

1.4 PROJECT SCOPE

The scope of this project focuses on the design and fabrication of a low-cost, three-wheeled personal transporter suitable for everyday short-distance mobility in Zambia, and will include the following key features:

- **PID Motor Control:** The transporter will use a digital Proportional-Integral-Derivative (PID) control system implemented on an Arduino Uno to ensure smooth acceleration, steady speed, and stable performance under varying loads and road conditions.
- **Brushless Hub Motors:** The vehicle will be powered by brushless hub motors, sized appropriately to carry a load of up to 70 kg while maintaining a speed range of 20–26 km/h, providing both efficiency and reliability.
- **Scavenged Lithium Battery:** A lithium battery from a hoverboard will be used as the main power source, ensuring affordability while providing sufficient energy for urban mobility and short-distance trips.
- **User-Friendly Design:** The transporter will be designed for ease of use, including intuitive speed control via a potentiometer and real-time speed feedback using a hall sensor embedded in the hub wheel.
- **Affordability:** Cost-effectiveness is a key focus of the project, with all components carefully selected to maintain a low-cost structure while ensuring functionality, reliability, and accessibility.
- **Lightweight and Portable:** The transporter will be lightweight and compact, making it easy for users to handle, transport, and store without difficulty.
- **Ergonomic and Safe:** The design will consider user comfort and safety, including appropriate handle bar height, floor clearance, and stability for different users.
- **Aesthetically Pleasing:** While functional, the transporter will also have a visually appealing design to encourage adoption and practical everyday use.

1.5 RELEVANCE OF THE PROJECT

This project provides a low-cost and reliable three-wheeled personal transporter that can improve mobility and independence for people in Zambia. It is especially useful for small delivery services and everyday transport, offering an affordable alternative to expensive vehicles. By reducing transportation costs, it makes personal mobility more accessible to more people.

The project also demonstrates how engineering and digital control technology can solve local transport challenges. The PID control system helps the transporter maintain smooth acceleration, steady speed, and stability under different loads, making it safer and more practical to use in urban and semi-urban areas.

The transporter's compact, lightweight, and user-friendly design makes it easy to handle and suitable for crowded streets. Using locally sourced or scavenged components, such as a

lithium battery from a hoverboard, shows that effective and affordable transport solutions can be built with available resources.

Overall, this project is important because it can improve everyday transportation, support small businesses, increase accessibility, and promote low-cost engineering innovations. There is a clear need for affordable and dependable personal transporters in Zambia.

CHAPTER 2: LITERATURE REVIEW

2.0 LITERATURE REVIEW INTRODUCTION

Personal transporters are compact electrically powered vehicles designed to carry individuals over short to medium distances. They provide a convenient and eco-friendly alternative to traditional transportation modes such as bicycles, motorcycles, and cars [14]. These vehicles are typically lightweight and easy to operate, making them particularly suitable for urban environments where traffic congestion and limited parking are common challenges. The evolution of personal transporters has included both two- and three-wheeled designs, each with specific advantages in terms of stability, manoeuvrability, and user safety [15,16]. In recent years, the increasing focus on sustainable mobility has spurred global interest in developing low-cost, energy-efficient personal transport solutions.

The uses of personal transporters are diverse, ranging from daily commuting and delivery services to recreational purposes [17]. Their affordability, compared to conventional vehicles, has made them attractive in developing countries and urban centers where personal vehicles are expensive or impractical. For instance, electric scooters can cost as little as ZMW 6,000–8,000 depending on specifications, while electric bikes may range between ZMW 30,000–40,000, and electric cars can exceed ZMW 1,200,000 [18,19]. Beyond individual mobility, personal transporters are increasingly used in logistics and last-mile delivery due to their compact size and energy efficiency, which reduces operational costs and environmental impact [20]. These economic and practical benefits make personal transporters a viable option for both personal and commercial applications.

A key factor in the functionality and safety of personal transporters is the technology used to control their operation. Modern transporters rely on a variety of electronic and mechanical systems to regulate speed, stability, and energy consumption [21]. Common control technologies include Proportional-Integral-Derivative (PID) controllers, microcontroller-based motor drivers, Hall-effect sensors for speed monitoring, and regenerative braking systems [22,23]. These systems ensure smooth acceleration, maintain steady speed under varying loads, and improve overall reliability. The integration of such control technologies is critical for making personal transporters practical for everyday use, particularly in regions with uneven terrain or variable load requirements.

These introductory paragraphs outline the three main themes that will be explored in detail throughout this literature review: the definition and evolution of personal transporters, their applications and associated costs, and the control technologies that enhance their performance. By establishing this context, the literature review provides a comprehensive framework for analyzing existing research and identifying opportunities for innovation in low-cost, reliable personal transporters.

2.1 COST ANALYSIS OF ELECTRIC TRANSPORT OPTIONS

Electric scooters (e-scooters) have become a cost-effective solution for short-distance urban mobility. They are lightweight, compact, and easy to store, making them ideal for navigating congested city streets while producing zero tailpipe emissions, contributing to cleaner urban environments. Bishop reported that the operating cost of electric scooters, including electricity and battery replacement, amounts to approximately £0.045 per kilometre, which is about ZMW 1.10 per kilometre at current exchange rates [18]. In India, electric two-wheelers cost roughly ₹1.48 per kilometre, equivalent to about ZMW 0.40 per kilometre, demonstrating their economic advantage over petrol-powered vehicles [19]. However, e-scooters have limitations, including limited range, reduced performance under heavy loads, and minimal protection in the event of accidents.

Electric bikes (e-bikes) provide advantages over e-scooters in terms of range and versatility. Most e-bikes can travel between 50–150 kilometres per charge, making them suitable for longer trips, and their pedal-assist feature allows users to continue manually when battery levels are low. McCarran reported that the average cost of an electric bike is about \$1,261, which translates to approximately ZMW 34,000, depending on the model and features [20]. Additionally, energy efficiency is high, with some models consuming as little as 0.05 kWh per 5 km, further reducing operational costs [21]. Despite these benefits, e-bikes are bulkier than e-scooters and have higher maintenance requirements, which may affect affordability and convenience for some users.

Electric cars, by contrast, offer greater comfort, safety, and range, making them suitable for long-distance travel and family transportation. Many electric cars can travel 300–500 kilometres on a single charge, and they feature enclosed cabins and advanced safety systems. The annual cost of fuelling an electric car is significantly lower than for petrol vehicles; the NRDC reports \$485 per year versus \$1,117 for gasoline cars, roughly ZMW 13,000 versus ZMW 30,000 annually [22]. However, high upfront purchase costs—averaging \$47,000 (≈ ZMW 1,250,000) and the larger environmental footprint of battery production make them less accessible for everyday short-distance use.

When comparing these transport modes, each presents clear strengths and limitations. E-scooters are highly affordable, portable, and environmentally friendly, making them ideal for short-distance urban commuting. E-bikes extend range and load capacity but are more expensive and less portable. Electric cars provide comfort, safety, and long-range capabilities but come at high cost and size limitations, reducing their practicality for urban travel. In the Zambian context, where affordability, compactness, and operational efficiency are crucial, e-scooters appear to be the most practical and sustainable choice for personal transportation [18–22].

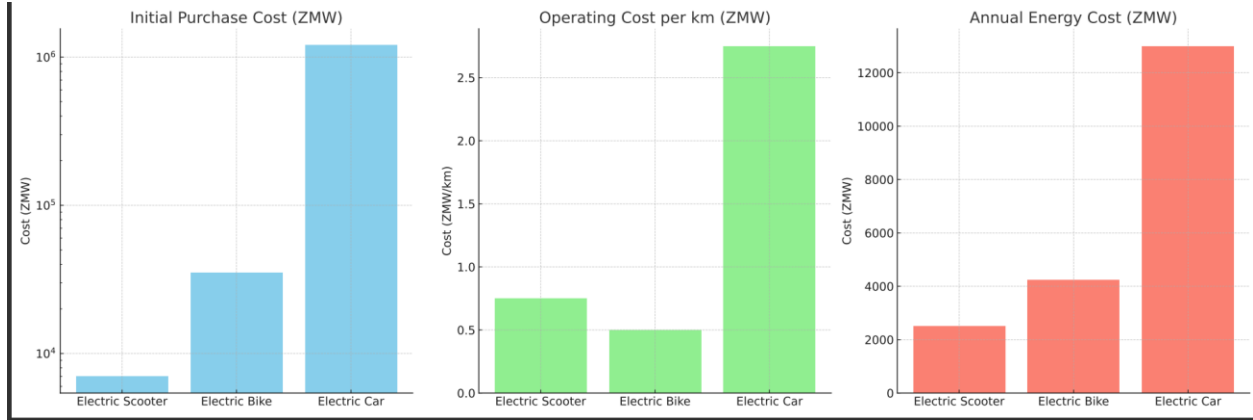


Figure 2.1 A Cost comparison chart

2.2 CONTROLLER STRATEGIES FOR BLDC MOTORS

2.2.1 OPEN-LOOP CONTROL

In an open-loop control strategy for Brushless DC (BLDC) motors, the motor speed is regulated by adjusting the duty cycle of a Pulse Width Modulated (PWM) signal without using any feedback from sensors. The duty cycle determines the average voltage applied to the motor according to:

$$V_{avg} = D \cdot V_{dc}$$

Where;

V_{avg} = Average voltage

D = Duty cycle

V_{dc} = Supply DC voltage

and the motor speed is assumed to be proportional to this voltage through the motor velocity constant:

$$\omega \approx K_v \cdot (D \cdot V_{dc})$$

ω = angular speed

K_v = Motor velocity constant (rad/s per V)

While this method is simple and low-cost, it suffers from major drawbacks due to the absence of feedback. Any change in load, such as increased rider weight or road slope, directly

reduces motor speed since the controller cannot self-correct. Similarly, disturbances like friction or bumps cause instability, and the motor often experiences jerky startups because rotor position is not monitored.

The governing equations of BLDC motors highlight these limitations. The electrical dynamics are expressed as:

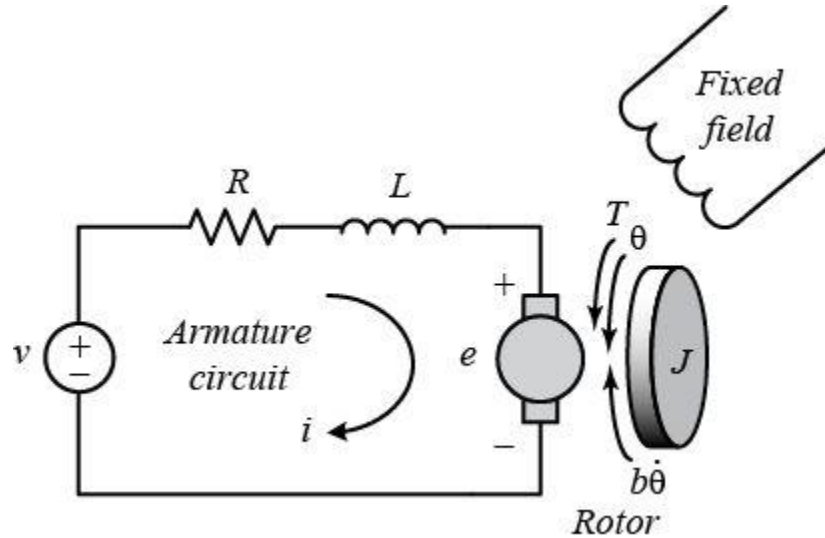


Figure 2. 2 Electro-Mechanical system model schematic

$$V = L \frac{di}{dt} + Ri + e$$

V= Applied phase voltage (V)

L=Motor winding Inductance(H)

$\frac{di}{dt}$ = Rateofchangeofcurrent(A/s)

R =Motor winding Resistance

I = Armature current

e = electromotive force

where the back-EMF (e) depends on motor speed, while the mechanical dynamics are given by:

$$T_e = J \frac{d\omega}{dt} + B\omega + T_L$$

J = Rotor inertia (kg·m²)

$$\frac{d\omega}{dt} = \text{Angular acceleration (rad/s}^2\text{)}$$

B = Viscous friction coefficient (N·m·s/rad)

T_L = Load torque (Nm)

B = viscous friction constant (N·m·s/rad)

In open-loop operation, the relationship between duty cycle, voltage, and speed is treated as linear, but in reality, variations in load torque cause significant deviations. This makes open-loop control inefficient and unreliable for applications such as personal transporters, thereby justifying the need for closed-loop strategies like PID control.

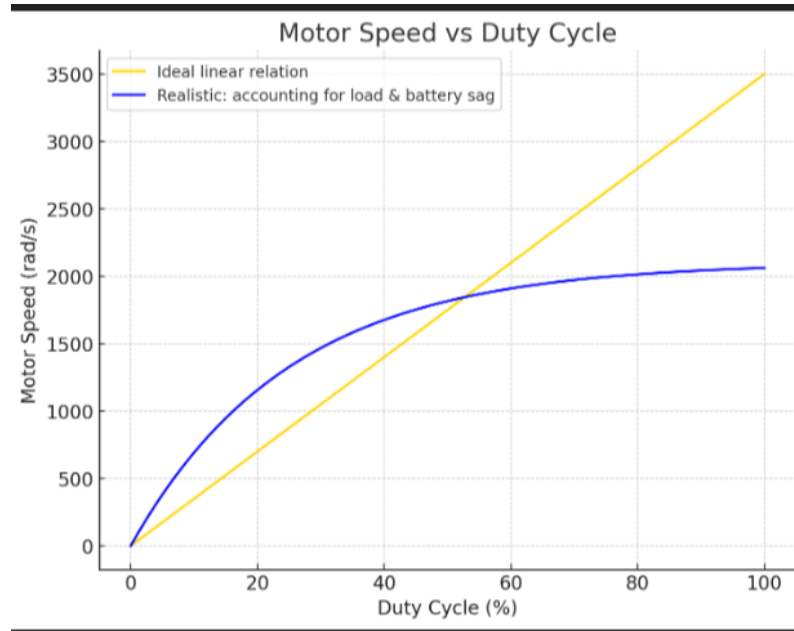


Figure 2. 3 Graph of Motor Speed vs Duty Cycle

Figure 2.3 illustrates the relationship between motor speed and duty cycle. In the ideal case, the motor speed is shown to increase linearly with duty cycle, suggesting a proportional relationship. However, the realistic scenario reveals that motor speed saturates as the duty cycle increases due to the combined effects of back electromotive force (back-EMF), load resistance, and mechanical friction. These non-linearities prevent the motor from achieving the theoretical maximum speed suggested by the ideal case. This observation is consistent with established electric motor behaviour, where external factors limit performance beyond a certain duty cycle threshold [47,48].

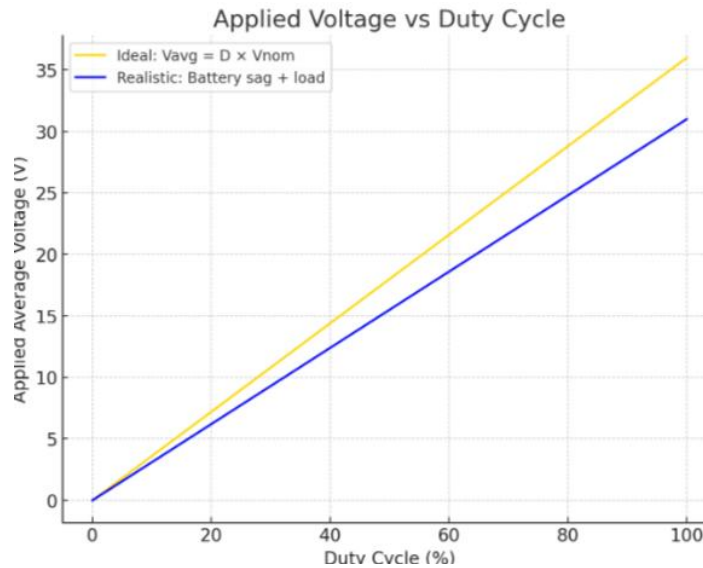


Figure 2. 4 Graph of Applied voltage vs Duty cycle

Figure 2.4 The second graph compares the applied voltage against the duty cycle. Under ideal conditions, applied voltage scales linearly with duty cycle, representing a perfect proportional relationship between the PWM signal and the effective voltage delivered to the motor. In practice, however, the applied voltage deviates from linearity because of battery voltage sag under load and internal resistance within the power system. These effects reduce the actual voltage supplied to the motor, especially under high current demands. Such deviations are commonly reported in electric drive systems where power sources cannot maintain constant voltage during heavy operation [47,48].

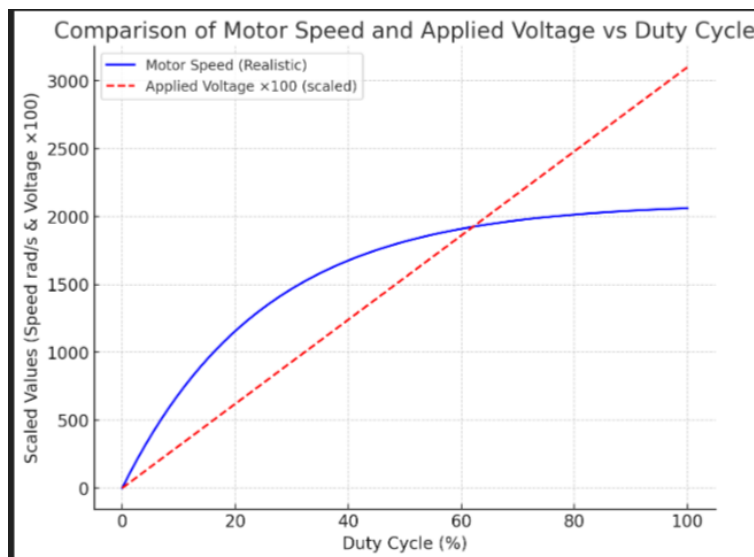


Figure 2. 5 Shows Comparison of motor speed and applied voltage vs Duty Cycle

The third graph presents both applied voltage and motor speed versus duty cycle on the same scale for comparison. The applied voltage remains closer to the ideal linear relationship but still exhibits minor drops due to battery sag, while the motor speed shows a more significant departure from the ideal because of the compounded influences of load torque, back-EMF, and frictional forces. This combined view highlights that while PWM duty cycle can effectively regulate applied voltage, the translation to motor speed is constrained by real-world factors. Such comparisons between theoretical and practical performance are emphasized in motor control literature to underline the need for feedback mechanisms like PID or PD controllers to ensure stable operation [47,48].

THE ROLE OF CONTROL STRATEGIES IN PERSONAL TRANSPORTER SYSTEMS

In designing a personal transporter, maintaining stability, speed regulation, and smooth user control is critical. The dynamics of electric motors, combined with external disturbances such as varying rider weights, road conditions, and battery voltage fluctuations, introduce challenges that require robust control strategies. Classical control methods such as Proportional-Integral (PI), Proportional-Derivative (PD), and Proportional-Integral-Derivative (PID) controllers have been extensively applied in motor drive systems because of their simplicity, reliability, and effectiveness. These controllers form the foundation for achieving desired performance indicators, including reduced steady-state error, improved rise time, and minimal overshoot. By reviewing PI, PD, and PID control strategies in the context of electric drive systems, this study establishes the theoretical framework necessary for selecting and implementing a suitable control method for the personal transporter. Their discussion is therefore essential to justify the control design choices made in later chapters of this work.

2.2.2 PI CONTROL

Proportional-Integral (PI) control is widely used in industrial drives and corrects both proportional and steady-state error [27]. However, without a derivative component, PI controllers struggle with overshoot and oscillations in dynamic systems such as BLDC transporters [28]. Its transfer function is given by;

$$G_{PI}(s) = K_P + \frac{K_i}{s}$$

Where K_P is the proportional control constant, K_I the integral control constant.

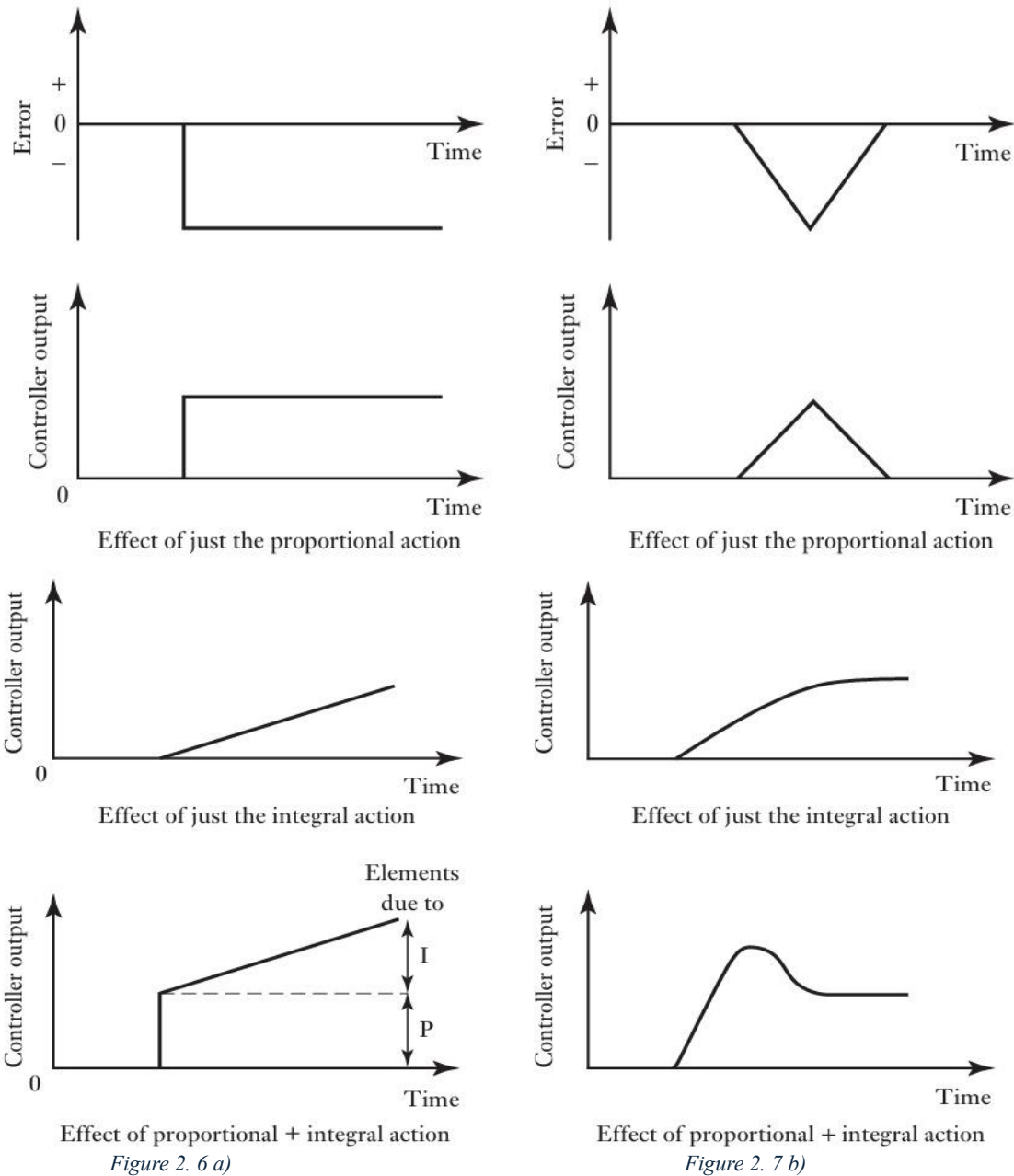


Figure 2.4(a) shows how the system reacts when there is an abrupt change to a constant error. The error gives rise to a proportional controller output which remains constant since the error does not change. There is then superimposed on this a steadily increasing controller output due to the integral action. Figure 2.4(b) shows the effects of the proportional action and the integral action if we create an error signal which is increased from the zero value and then decreased back to it again. With proportional action alone the controller mirrors the change and ends up back at its original set point value. With the integral action the controller output increases in proportion to the way the area under the error–time graph increases and since,

even when the error has reverted back to zero, there is still a value for the area, there is a change in controller output which continues after the error has ceased. [37]

2.2.3 PID CONTROL

Proportional–Integral–Derivative (PID) control extends PI by anticipating error trends, reducing overshoot and improving transient response. Recent experiments demonstrate that PID improves speed regulation, smoother acceleration, and stability under nonlinear load conditions in BLDC motors [28,29]. Having chosen PID as our focus in this project most of its content is discussed in chapter 3; Methodology.

$$G_{PID}(s) = K_P + \frac{K_I}{s} + K_D$$

Where; K_D is the Derivate control constant, K_P is the proportional control constant, K_I the integral control constant.

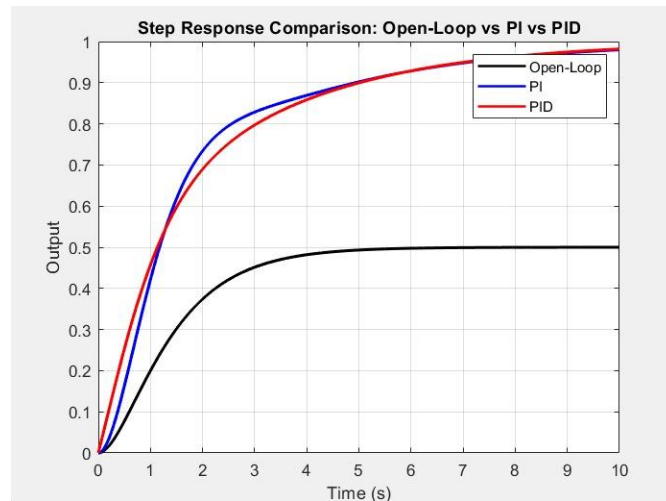


Figure 2. 8 Step Response Comparison: Open-Loop vs PI vs PID

The graph (code found in *Appendix C* compares the step response of three control systems: Open-Loop, PI (Proportional-Integral), and PID (Proportional-Integral-Derivative). Based on the graph, a comprehensive comparison of the three control systems justifies the use of PID control. The Open-Loop system has a steady-state output of approximately 0.5, which is less than the desired output of 1. In contrast, both PI and PID controllers achieve a steady-state output close to 1, indicating better tracking of the setpoint. The PID controller shows a slightly faster response compared to the PI controller, with both having minimal overshoot and settling to the steady-state output.

PID control generally offers better performance due to the addition of the derivative term, which helps in predicting and correcting future errors, thus improving stability and reducing

overshoot. PID controllers are widely used for their effectiveness in a variety of control applications [38]. PID controllers are effective in reducing steady-state error (due to the integral term) and improving transient response (due to the proportional and derivative terms) [39]. The derivative term in PID control adds predictive capability, helping to dampen oscillations and improve stability [40].

2.2.4 HYBRID AND ADVANCED CONTROLLERS

Recent research (2020–2025) explores hybrid solutions such as fuzzy-PID, adaptive PID, and neuro-fuzzy approaches. For example, Kroičs and Būmanis (2024) developed an adaptive fuzzy-PID controller that reduced torque ripple and improved speed regulation [29]. Similarly, Patel et al. (2022) proposed an adaptive fuzzy PID for BLDC drives, achieving lower overshoot and better load compensation than classical PID [30]. More advanced techniques such as Model Predictive Control (MPC) and ANFIS-based controllers also show promise [31,32], but their computational complexity limits practical use in low-cost devices.

2.3 JUSTIFICATION FOR PID SELECTION

Although advanced controllers may achieve superior performance, their computational and cost demands make them impractical for low-cost Zambian transporters. PID offers the best balance between performance, affordability, and simplicity. It is:

- **Proven and widely adopted:** PID is the most implemented industrial controller [27,29].
- **Resource efficient:** It can run on low-cost microcontrollers like Arduino [35].
- **Balanced:** More robust than PI, but less complex than fuzzy or predictive methods.
- **Demonstrated in practice:** Arduino-based PID hoverboard motor control has achieved reliable operation under varying load conditions [35].

Thus, PID is selected for this project, with scope for adaptive enhancements if future upgrades are feasible.

2.4 SENSOR TECHNOLOGIES FOR FEEDBACK

Accurate and reliable feedback is essential for closed-loop control in personal transporters, as it enables the system to monitor performance variables such as speed, position, and orientation, and then adjust motor input accordingly. Without proper feedback, controllers like Proportional Integral Derivative cannot effectively compensate for disturbances caused by rider weight, terrain variation, or sudden load changes. Sensor technologies therefore form

the backbone of intelligent motor control, ensuring smooth acceleration, stable cruising, and safe operation.

Different sensors provide unique advantages depending on the application. Hall-effect sensors, typically embedded in hub motors, deliver rotor position and speed information with high reliability and low cost. Inertial Measurement Units (IMUs), which combine accelerometers and gyroscopes, allow monitoring of angular velocity and inclination—useful for disturbance rejection and slope detection. Rotary encoders, though more expensive, offer superior precision and are widely used in applications demanding fine resolution of speed and position. Selecting the appropriate sensor or combination of sensors is a critical design consideration, balancing factors such as accuracy, repeatability, cost, and environmental robustness.

This section reviews the key sensor technologies relevant to personal transporters, highlighting their principles of operation, applications in feedback control, and trade-offs in performance versus cost.

2.4.1 HALL-EFFECT SENSOR

Hall-effect sensors remain the default in hoverboard motors for rotor position feedback [24]. Alternative sensors include optical encoders, which offer higher resolution but are fragile outdoors [36]. IMUs, such as the MPU6050, are increasingly used for additional feedback on inclination and disturbance rejection [33]. Sensor fusion with Kalman filtering further improves accuracy [34]. These developments make IMUs highly relevant for transporters operating on uneven terrain.



Figure 2. 9 Shows Hall Sensors (H1, H2, H3) in a Hub motor Wheel

Hall effect sensors are transducers that detect the presence and magnitude of a magnetic field and convert it into a measurable electrical signal. In hub motors, they are typically used to determine rotor position and motor speed.

In brushless DC (BLDC) hub motors, Hall effect sensors are positioned inside the motor casing to sense the magnetic field from the permanent magnets. The output is a square-wave signal whose frequency is directly proportional to the rotational speed of the motor. The speed can be expressed as:

$$\omega = \frac{2\pi f}{N_p}$$

Where f is the signal frequency, and N_p is the number of pole pairs of the motor.

In the proposed transporter, Hall sensors provide feedback to the Arduino for closed-loop control of motor speed.

Hall sensors are accurate for speed estimation at moderate speeds, with typical resolution up to one rotation per pole-pair transition. They offer high repeatability since they are digital devices, but their accuracy may degrade at very high speeds due to missed pulses. Availability is high, as they are built into almost all hoverboard hub motors, reducing cost and complexity.

2.4.2 INERTIAL MEASUREMENT UNITS (MPU6050)

An IMU is an electronic device that measures acceleration and angular velocity, often combining a 3-axis accelerometer and gyroscope [37].

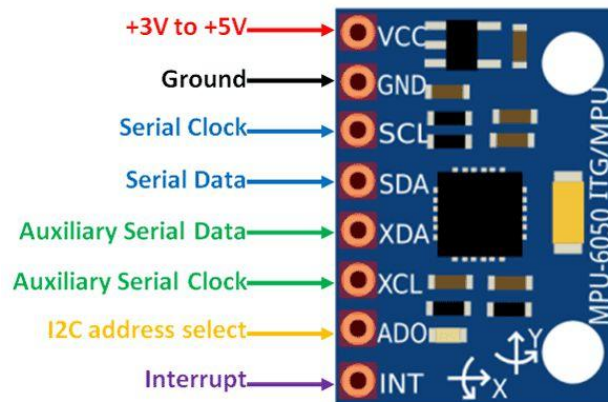


Figure 2. 10 Shows the MPU 6050

IMUs are critical in balancing systems such as hoverboards and Segways, where they provide orientation and tilt angle. For a three-wheeled transporter (with natural balance), IMUs may be used for monitoring slope or rider stability rather than balance control.

The IMU output can be used to estimate tilt angle through sensor fusion

$$\theta \approx \arctan\left(\frac{\theta_y}{\theta_z}\right)$$

Where θ_y and θ_z are accelerometer readings.

IMUs offer high accuracy and sensitivity, with errors mainly due to drift in gyroscope signals. They are repeatable but require filtering to achieve stability. Their availability is excellent, as low-cost IMUs like MPU6050 are widely sold for Arduino applications.

2.5.3 ROTARY ENCODERS

A rotary encoder is an electromechanical device that converts angular position or motion of a shaft into digital or analogue signals. It is widely used for precise measurement of rotational speed and position [37].

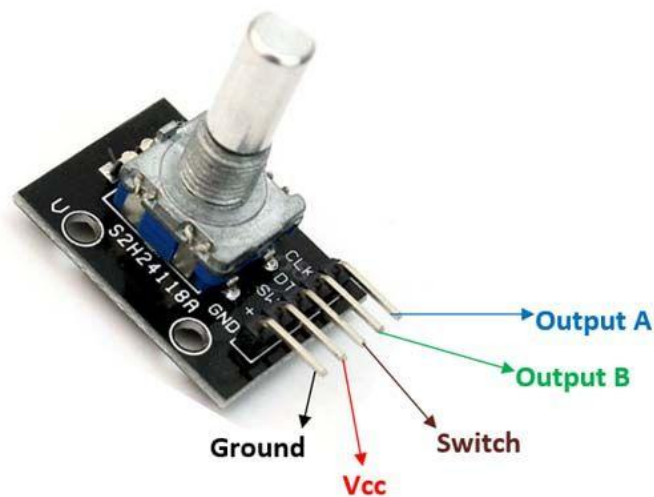


Figure 2. 11 A Shows a Rotary Encoder

Rotary encoders work by generating pulses as the shaft rotates. The frequency of these pulses corresponds to the angular speed, while counting the pulses gives position. The relationship is given by:

$$\omega = \frac{2\pi f}{N_{ppr}}$$

where f is the output pulse frequency, and N_{ppr} is the number of pulses per revolution of the encoder. [37]

For the personal transporter, rotary encoders could be used as an alternative or complement to Hall effect sensors to measure motor speed more accurately. Incremental encoders provide pulse counts relative to movement, while absolute encoders provide the exact shaft position.

Rotary encoders are highly accurate, with resolutions often reaching thousands of pulses per revolution, making them more precise than Hall sensors for speed control. They also offer excellent repeatability. However, they can be more costly and are not always integrated into low-cost hub motors, making them less readily available compared to Hall sensors. [37]

2.6 SAFETY AND POWER CONSIDERATIONS

Safety remains critical in BLDC transporters. Studies emphasize integrating fuses, resettable breakers, and battery management systems to protect against high currents [36]. For Zambia's market, ensuring affordability while embedding essential protective systems is key to long-term adoption.

2.7 EMERGING TRENDS

Between 2020 and 2025, research has focused on hybrid controllers, adaptive PID tuning, and sensor fusion. These trends show strong potential but are not yet widely accessible in low-cost transporters. PID remains the most feasible short-term solution, while adaptive fuzzy-PID and MPC approaches represent future directions [29–32,34].

2.8 PID AS THE PREFERRED CONTROLLER

From the reviewed literature, it is evident that Proportional–Integral–Derivative (PID) control provides the most balanced solution for personal transporters when compared with open-loop, PI, or more advanced hybrid controllers. While open-loop and PI controllers are simple, they struggle with disturbance rejection, overshoot, and maintaining stability under variable loads [27,28]. Advanced approaches such as fuzzy-PID and model predictive control show improved performance [29–32], but their computational complexity and high cost limit their feasibility in low-cost transporters. PID control, on the other hand, offers proven robustness, reliable stability, and resource efficiency, making it implementable on low-cost microcontrollers such as Arduino [35]. Furthermore, PID has been extensively validated in similar BLDC motor drive applications, demonstrating smooth acceleration, steady speed regulation, and effective load compensation [28,29,38–40]. Thus, PID provides the best trade-off between cost, simplicity, and effectiveness, aligning with the design priorities of affordable transport solutions for Zambia.

2.9 STABILITY OF BLDC MOTOR SYSTEMS

The literature also confirms that Brushless DC (BLDC) motors, when coupled with appropriate feedback sensors and closed-loop control, offer reliable stability and efficiency for personal transport applications. BLDC motors eliminate mechanical brushes, reducing wear and maintenance while providing a high torque-to-weight ratio [23,24]. However, their sensitivity to load variations requires closed-loop feedback for consistent speed and smooth operation [25]. Studies highlight that incorporating Hall-effect sensors, IMUs, or rotary encoders ensures precise rotor position and speed measurement, which is critical for stable performance [24,33–36]. Experimental results further demonstrate that BLDC drives, under PID control, achieve enhanced stability by reducing oscillations and maintaining set speed despite changes in rider weight or terrain [28,29]. This evidence confirms that BLDC motors, supported by closed-loop control and sensor feedback, provide a stable and dependable foundation for personal transporter systems.

2.10 CONCLUSION

This literature review shows that personal transporters are an affordable, eco-friendly alternative to conventional vehicles, with e-scooters being the most practical option for Zambia. BLDC motors are the preferred drive system due to their efficiency, but require closed-loop control to achieve stability under variable loads. Among control strategies, PID offers the most balanced trade-off between cost, simplicity, and effectiveness, making it ideal for the current project. Emerging adaptive and hybrid controllers hold promise, but PID remains the most implementable solution in Zambia's cost-sensitive context [23–36].

CHAPTER 3 METHODOLOGY

In this chapter, we give detailed procedures and design steps followed in the development of the 3-wheeled personal transporter system. It includes the mechanical system design and software integration strategy, control system implementation using Proportional-Integral Derivative (PID) logic, electronic system which includes sensor interfacing, and feedback mechanisms.

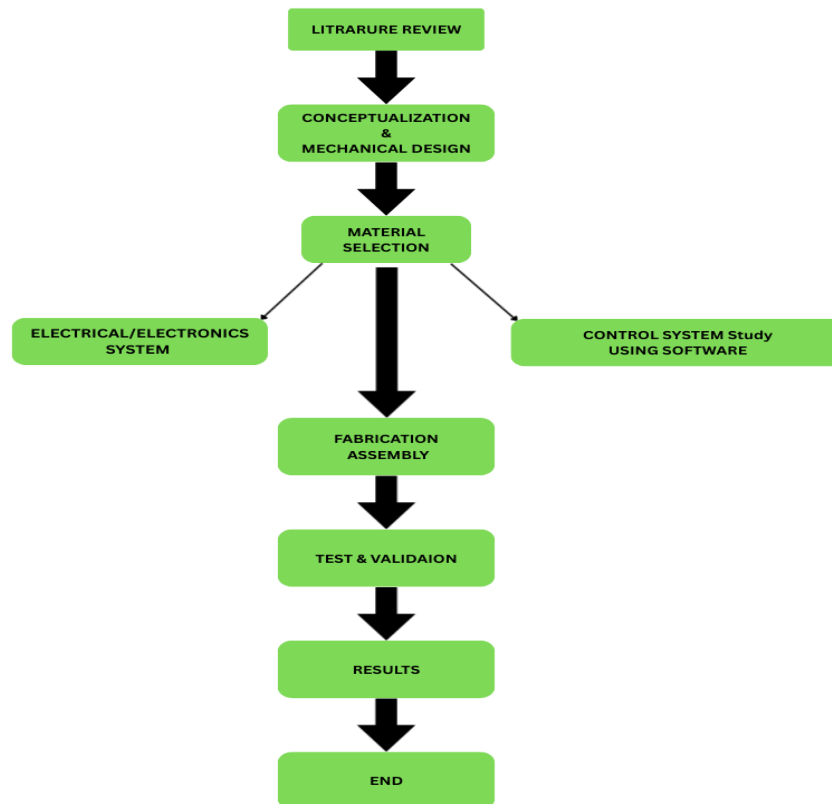


Figure 3. 1 Shows the Methodology

3.1 SYSTEM OVERVIEW

The personal transporter system consists of two brushless DC (BLDC) hub motors mounted on the rear wheels for propulsion, One non-motorized front wheel for manual steering, an Arduino Uno as the main controller, a ZS-X11H V2 BLDC motor controller for power delivery to the motors, a hall effect sensor for speed feedback, a voltage divider circuit to monitor battery voltage, and a 16x2 I2C LCD for real-time display of speed and battery level.

3.2 MECHANICAL DESIGN

The transporter uses a triangular wheel setup with two 16.5 cm (6.5-inch) motorized hoverboard wheels at the back and a single bicycle wheel at the front for steering. Its metal frame is built to carry both the rider and the electronics, while leaving enough room to mount the battery pack, controller, and wiring neatly. The frame is kept low to the ground to improve balance and stability when riding. For steering, the front wheel connects to a handlebar for simple, intuitive control, but the design also leaves room to add a steering linkage in future upgrades.

3.2.1 CONCEPTUAL DESIGN

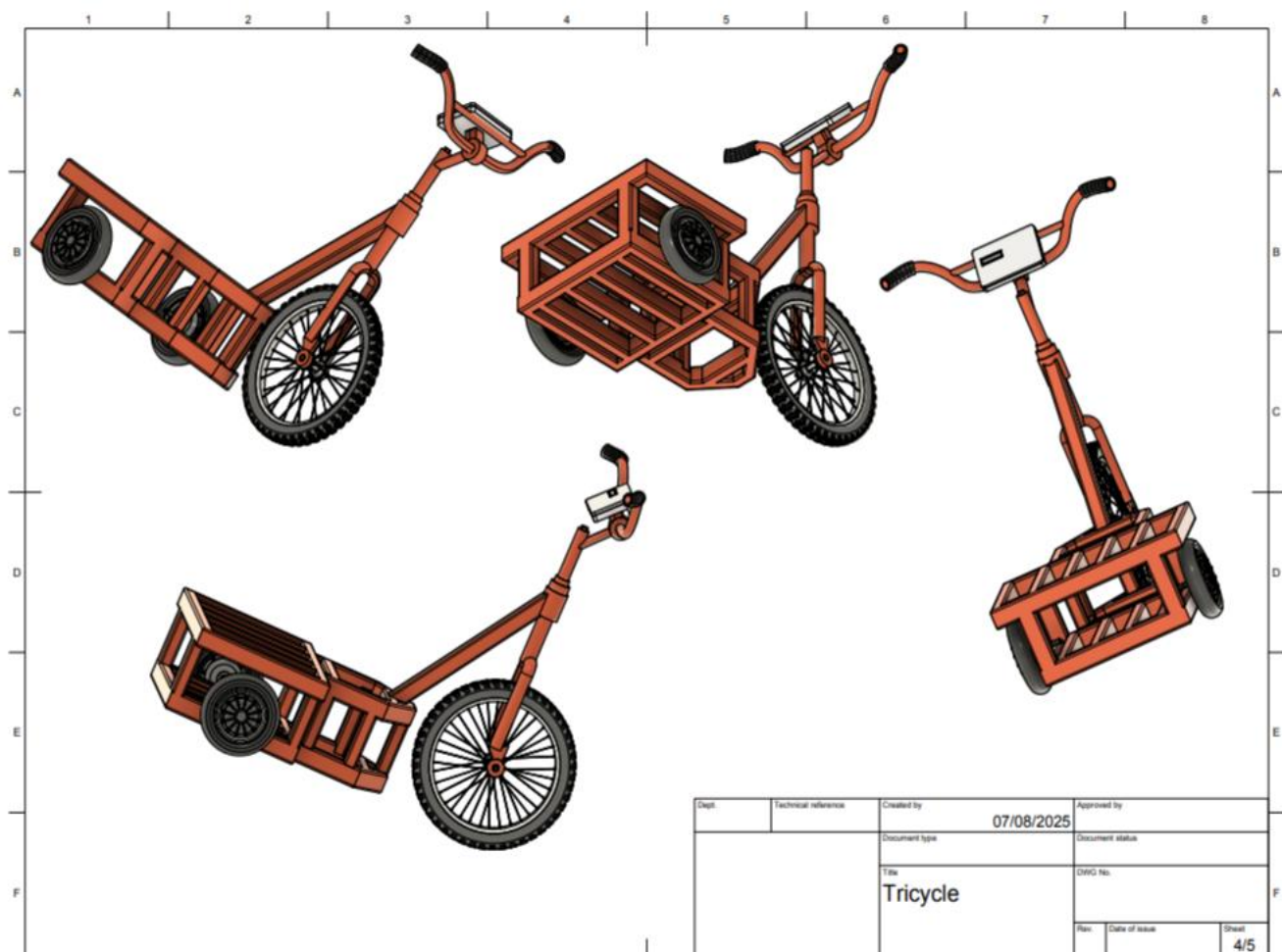


Figure 3. 1 Shows the Mechanical Conceptual design



Figure 3. 2 shows the isometric view

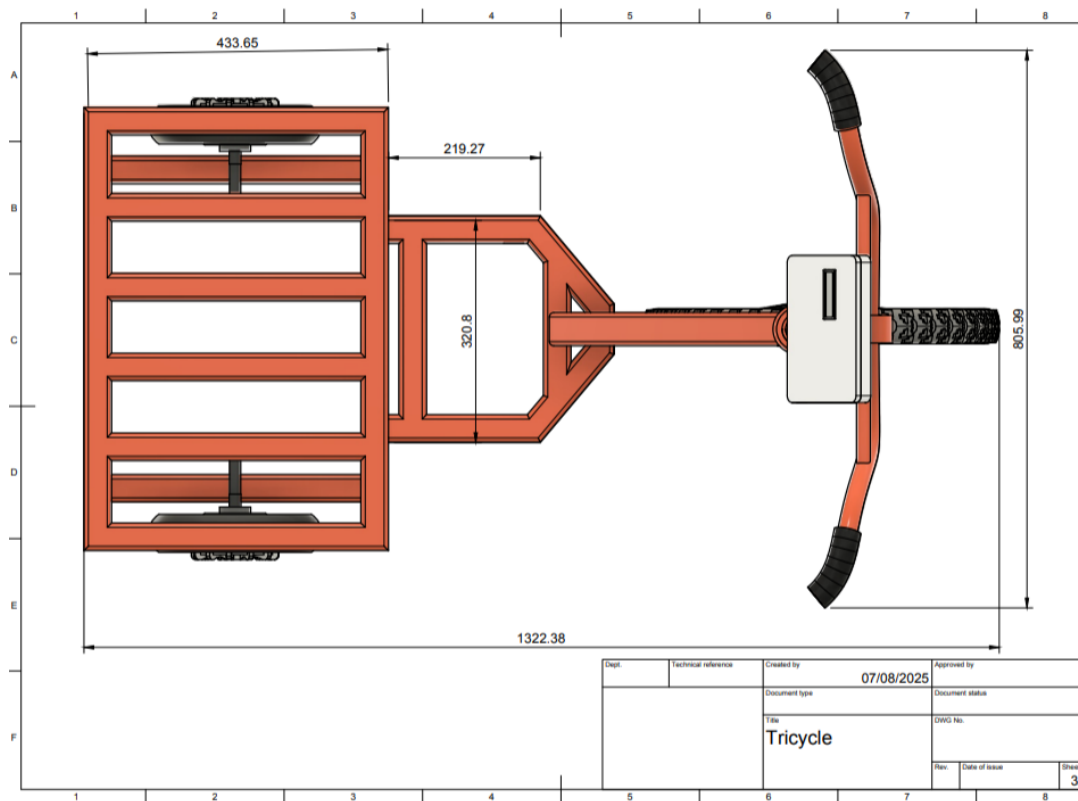


Figure 3. 2 Shows the Top View (length in mm)

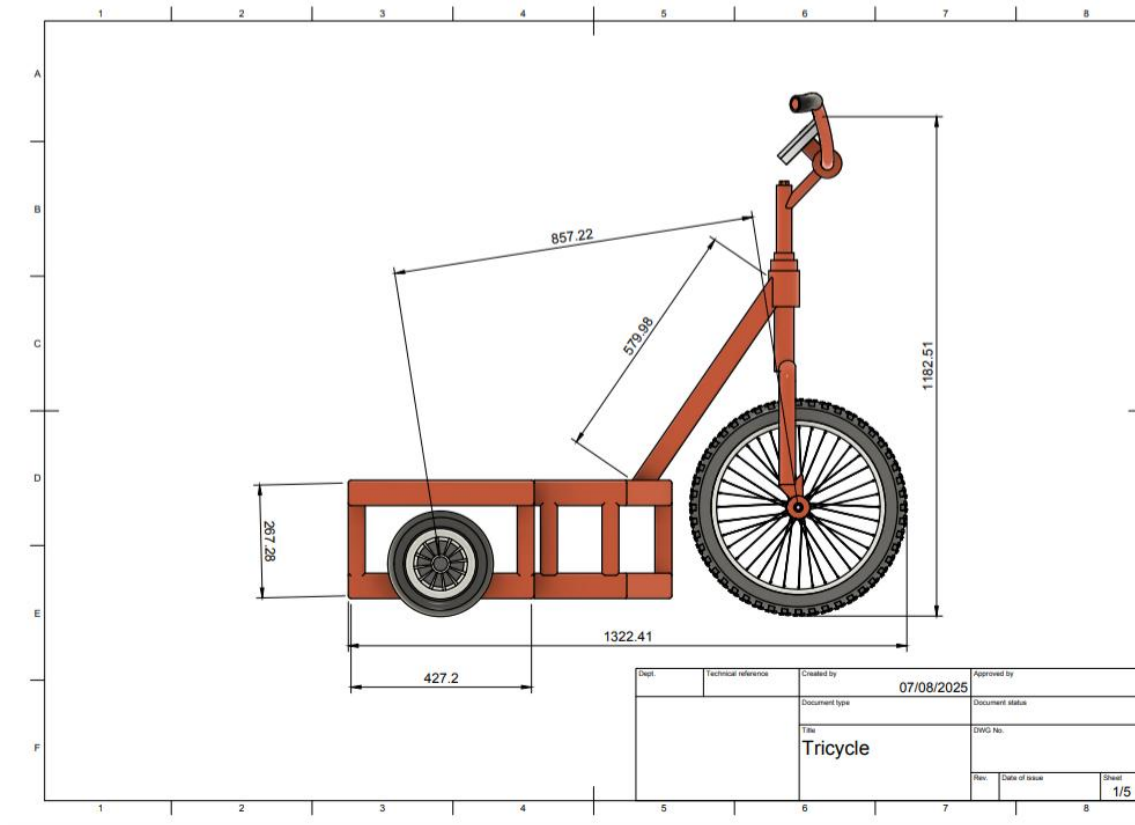


Figure 3. 3 Shows the Side View (length in mm)

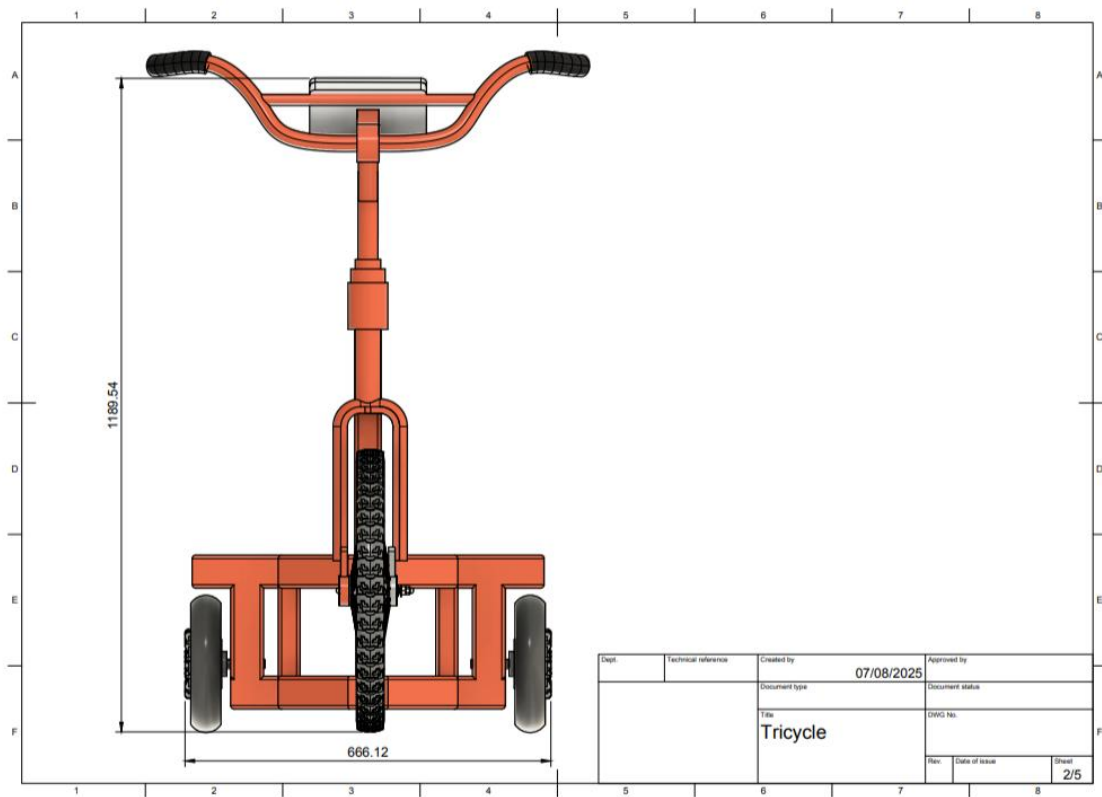


Figure 3. 4 Shows the Front View (length in mm)

3.2.2 DESIGN PRINCIPLES.

One of our key features in the design was that we wanted the rider to have an inherent or intuitive way to ride the Scooter. Simplicity sometimes is all you need, so rather than have multiple configurations or buttons that might make it difficult hard for user to turn or ride forward why not have a steering mechanism like a bicycle. Simple, precise, enjoyable all too familiar feeling. The bearing, bearing cups, and other related parts that connect a bicycle fork to the frame's head tube and allow steering are collectively called the **headset**.

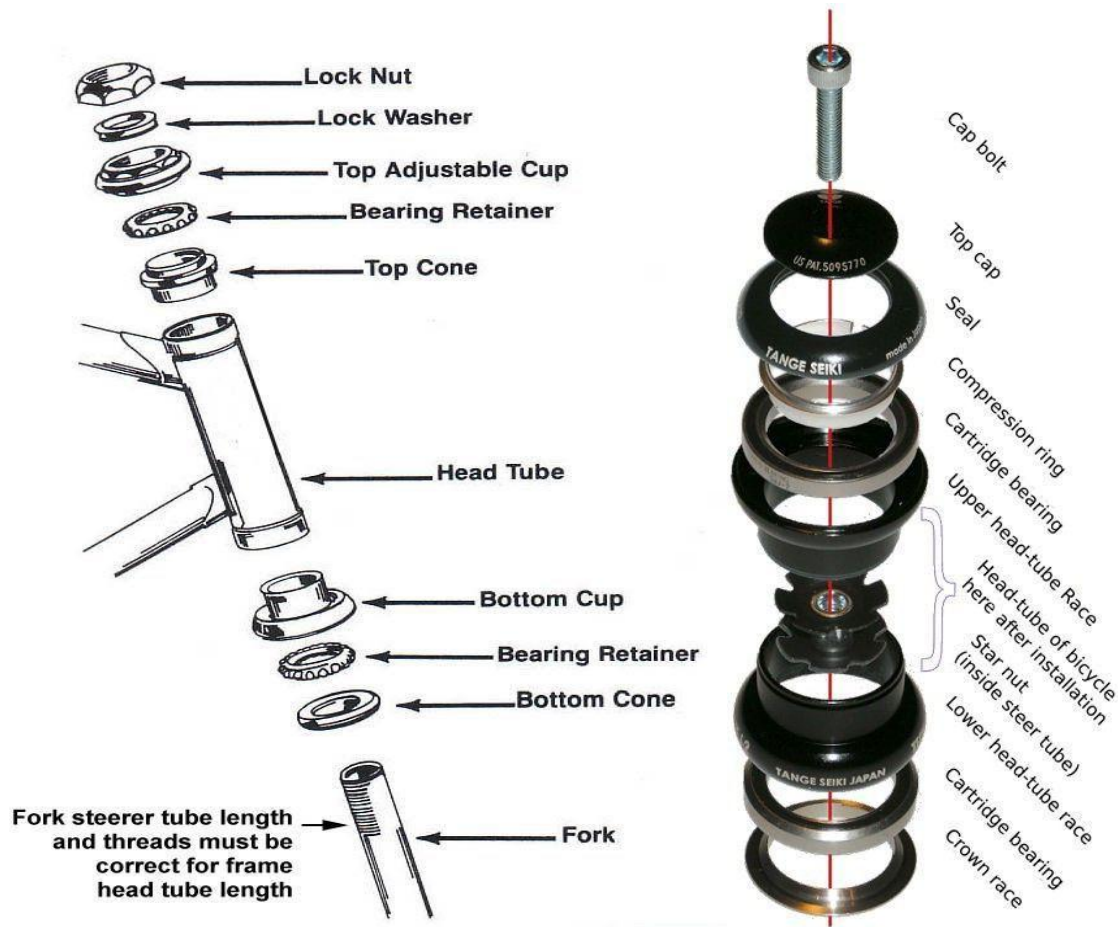


Figure 3. 5 Shows the Headset

To provide effective steering for the personal transporter, a bicycle-style handlebar system was implemented. The design initially utilized a fork salvaged from a small bicycle; however, standard tubes available on the market were not sufficiently long to accommodate a standing rider. To overcome this limitation, a custom solution was developed using a 1-meter steel tube with an internal diameter of 28 mm and an external diameter of 30 mm, which allowed for secure fitting of the bearing cups and proper integration with the fork. Additionally, a 90 cm M8 threaded rod was employed to ensure robust mechanical linkage, enabling smooth turning and reliable control of the personal transporter despite the extended steering column

length. This approach ensured both functionality and structural integrity, while addressing the ergonomic requirement of a standing rider.

3.2.3 STRESS ANALYSIS

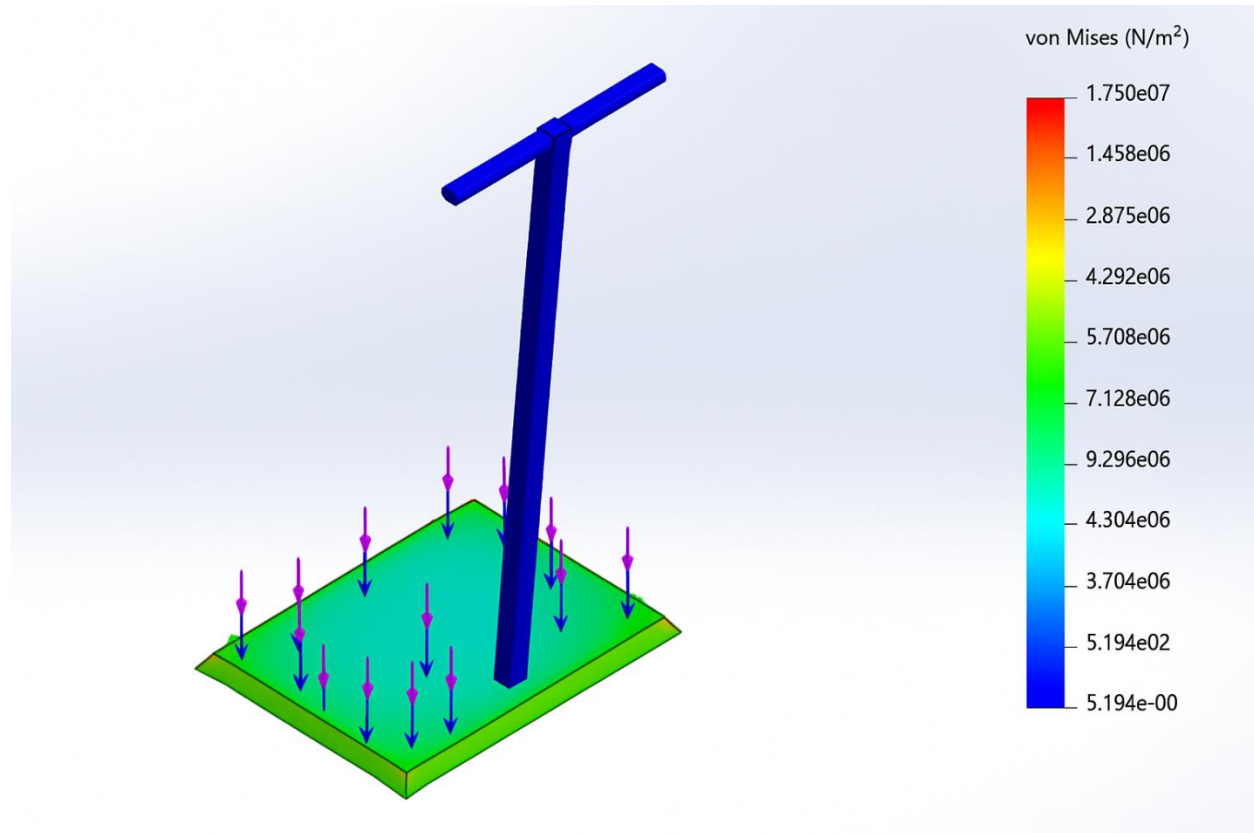


Figure 3. 6 Stress analysis on standing platform

A finite element analysis (FEA) was performed in SolidWorks to evaluate the structural integrity of the personal transporter's steering column and handlebar assembly under expected loading. Vertical loads, representing the rider's weight and operational forces, were applied to the base plate while the underside of the base was fully constrained to simulate ground support. The computed von Mises stress distribution shows a maximum stress of approximately $1.75 \times 10^7 \text{ N/m}^2$ (17.5 MPa) with minimum values near 5.19 N/m^2 (0.0052 MPa); the highest stresses are localized at the junction between the long steering tube and the handlebar. Comparing the maximum stress to typical structural materials (for example, mild steel yield $\approx 250 \text{ MPa}$) yields a conservative factor of safety ($\text{FOS} \approx 14$), indicating the design comfortably resists the applied loads. Prior to final reporting we verified unit consistency, refined the mesh near stress concentrations, and confirmed equilibrium of reaction forces in the model to ensure the results are physically meaningful.

3.3 MOTOR DESIGN CALCULATIONS AND SELECTION JUSTIFICATION

The motor selection process is central to ensuring the reliable operation of a personal transporter. The chosen motors must provide sufficient torque and power to propel both the rider and the transporter while meeting the desired performance targets. This section presents the design calculations to justify the selection of a **6.5-inch hub motor (350 W, 36 V)**, while also comparing it with larger hub motor options.

Step 1: Define the Load Requirements

The transporter is designed to carry:

- Rider weight (W_1): 70 kg
- Transporter weight (W_2): 7 kg
- Total system mass (m):

$$m = W_1 + W_2 = 70 + 7 = 77 \text{ kg}$$

The total gravitational force acting is:

$$F = m \times g = 77 \times 9.81 = 755.37 \text{ N}$$

Since two hub motors are used, each motor must support approximately half this load:

$$F_{\text{motor}} = \frac{F}{2} = \frac{755.37}{2} = 377.7 \text{ N}$$

Step 2: Required Torque

The torque requirement per motor is given by:

$$T = F_{\text{motor}} \times r$$

For a **6.5-inch (0.165 m) diameter wheel**,

$$r = \frac{0.165}{2} = 0.0825 \text{ m}$$

$$T = F_{\text{motor}} \times r = 377.7 \times 0.0825 = 31.1 \text{ Nm}$$

Thus, each motor must deliver at least **31.1 Nm of torque** to move the load

Step 3: Motor Capability

According to the supplier specifications [41], a 6.5-inch, 36 V, 350 W BLDC hub motor provides:

- **Rated torque:** ~35–40 Nm
- **Rated speed:** ~25–28 km/h
- **Maximum speed requirement for project:** 26 km/h

This confirms that the selected motor meets the torque requirement (≥ 31.1 Nm) and satisfies the target speed.

Step 4: Power Requirement Check

The theoretical power demand is calculated from:

$$P = F \times v$$

Where:

- $F = 755.37$ N
- $v = 26$ km/h = 7.22 m/s

$$P = 755.37 \times 7.22 = 5450 \text{ W (total for system)}$$

However, this value represents the instantaneous peak power required to accelerate the entire system from rest, not the continuous demand. In practice:

- Motors only need to overcome rolling resistance and provide additional torque during acceleration.
- With a rolling resistance coefficient $C_{rr} = 0.015$

$$F_{rr} = C_{rr} \times F = 0.015 \times 755.37 = 11.3 \text{ N}$$

$$P_{rr} = F_{rr} \times v = 11.3 \times 7.22 = 81.6 \text{ W}$$

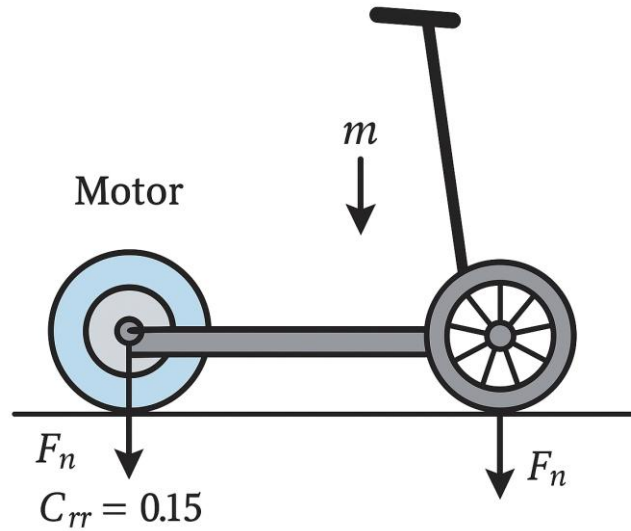


Figure 2. 12 Free body diagram of a cart showing motor force, rolling resistance and mass

Thus, under steady cruising, the motors consume **much less than their rated 350 W each**, leaving sufficient headroom for acceleration and slope climbing.

Step 5: Comparison with Larger Motors

Wheel Size	Typical Rated Power	Torque Capacity	Remarks
6.5 inch	350 W	35–40 Nm	Adequate for 77 kg load
8 inch	500 W	45–50 Nm	More torque but higher cost and bulk
10 inch	800–1000 W	70+ Nm	Overdesigned for current application

While larger motors can provide more torque, they introduce unnecessary **cost, weight, and complexity**. The **6.5-inch motor** offers a balance of affordability, availability, and sufficient performance for the intended rider weight and transporter size.

Conclusion

From the above analysis, it is evident that the **6.5-inch, 350 W hub motor** provides adequate torque (≥ 35 Nm) and speed (26 km/h) for transporting a 70 kg rider and a 7 kg vehicle. Larger hub motors, though more powerful, would result in unnecessary expense and additional mass. Therefore, the 6.5-inch motor is selected as the most efficient and practical choice for this design.

3.4 ELECTRICAL/ ELECTRONIC SYSTEM DESIGN

3.4.1 POWER SUPPLY

The transporter is powered by a 36 V, 2.2 Ah lithium-ion battery pack. Lithium-ion batteries are widely used in hoverboards and small electric vehicles due to their high energy density, light weight, and rechargeability [42].



Figure 3. 1 Shows a Lithium-Ion battery

(I) JUSTIFICATION & SIZING METHOD

The motor drivers and hub motors operate at ~36 V. Using a 36 V battery avoids heavy DC-DC conversion and matches driver input range, to size capacity we use:

$$Runtime(Hours) = \frac{BatteryEnergy(Wh)}{Averagesystemelectricalpower(W)}$$

where Battery energy (Wh) = Nominal voltage $\times$ Ah

Estimate average electrical power: Conservative cruising + real usage average around **400–700 W** combined for both motors (depends on rider, slope, acceleration)

The system uses two hub motors, each rated at 350 W. Depending on usage, the power demand can be summarized as:

- Single motor operation (light load): 350 W

- Dual motor operation (full load): 700 W

Using the energy capacity of the prototype battery:

- At 350 W load:
Runtime = $79.2 \text{ Wh} / 350 \text{ W} \approx 0.23 \text{ h} \approx 14 \text{ minutes}$
- At 700 W load:
Runtime = $79.2 \text{ Wh} / 700 \text{ W} \approx 0.11 \text{ h} \approx 7 \text{ minutes}$

It is important to note that this runtime is applicable if the device is continuously used at full load, in reality though it is rare that something is continuously used at full load. This means that the system can run longer than the 7 minutes calculated.

Battery capacity	Energy (Wh)	Runtime @ 700 W
2.2 Ah (pack of choice)	79.2 Wh	7 min
10 Ah (recommended)	360 Wh	30.6 min
15 Ah (recommended)	540 Wh	46 min

(II) JUSTIFICATION OF USE OF 2.2 AH

Although a 2.2 Ah pack provides very short runtime ($\approx 7\text{--}14$ minutes depending on load) compared to recommended 10–15 Ah packs, it was selected because it was cheap, readily available locally, and adequate for initial prototype testing. For final design or practical use, a higher-capacity battery would be required to achieve acceptable runtimes.

(III) COMPARISON WITH STANDARD HOVERBOARD BATTERIES:

According to *The Battery Tips* [43], high-quality hoverboard batteries typically range from 36 V, 4 Ah (144 Wh) to 36 V, 4.4 Ah ($\approx 158 \text{ Wh}$), providing ride times of 30–60 minutes under moderate loads. These packs also include advanced safety features, such as overcharge, short-circuit, and thermal protections, and can maintain 85% of their capacity after 1500 charge cycles. In contrast, the 2.2 Ah pack provides only about 8 minutes runtime, lacks robust cycle life, and may experience voltage sag under high current demand.

(IV) IMPLICATION FOR DESIGN

While the chosen 2.2 Ah battery is suitable for proof-of-concept, the final design will require a higher-capacity lithium-ion pack ($\geq 10 \text{ Ah}$) to achieve practical runtimes and safe current

delivery. Selecting a reputable brand with UL 2272 certification ensures not only longer ride distances but also user safety and compliance with international standards [44].

3.4.2 MOTOR CONTROLLER: ZS-X11H V2

Component: ZS-X11H brushless DC motor driver module, 400 W, 9–60 V, rated current 16 A (continuous), peak 20 A. [42]

(I) *WHY THIS DRIVER IS CHOSEN*

- **Voltage compatibility:** Working voltage 9–60 V covers the 36 V pack with margin.
- **Power & current:** Rated up to 400 W; suitable for many hoverboard/ hub motors and small transporters. The built-in ratings (16 A / 20 A peak) match many hub motor needs but final motor choice must be checked against these current ratings.
- **Hall-sensor support:** Driver accepts Hall effect sensors, simplifying commutation interface for hub motors that include Hall sensors.
- **Protection features:** Built-in overcurrent protection is useful as a first line of defense.
- **Small form factor & cost:** Compact PCB (63×45×31 mm) is appropriate for a DIY transporter with limited mounting space.
- **Availability:** Widely available and commonly used in hoverboard DIY projects so community support and wiring examples exist.

(II) *CAVEATS & SIZING NOTES*

- The ZS-X11H's 16 A continuous rating implies a maximum continuous motor electrical power near $P=V \times I=36 \times 16=576$ W per controller (ignoring losses). Two controllers (one per motor) should be used for dual-hub drives.
- If the anticipated continuous motor power per wheel exceeds what the driver can support (e.g., >500 W continuous), choose a higher-rated driver or ensure motor continuous draw below driver rating.
- Always use an external fuse on the battery line even with driver protection.

(III) *WIRING WITH EXTERNAL POTENTIOMETER AND DRIVERS*

The ZS-X11H V2 motor driver can be operated in a standalone configuration using an external potentiometer for throttle control. In this setup, the potentiometer is directly connected to the analog control input of the driver, where the variation in resistance corresponds to a proportional change in motor speed. The driver interprets the analog voltage

generated by the potentiometer and adjusts the output duty cycle to the motor, thereby controlling speed. This method is simple, cost-effective, and ideal for initial system validation since it does not require any external microcontroller or software tuning. It also allows the operator to manually vary the speed of the hub motor while the driver handles power switching and protection internally.

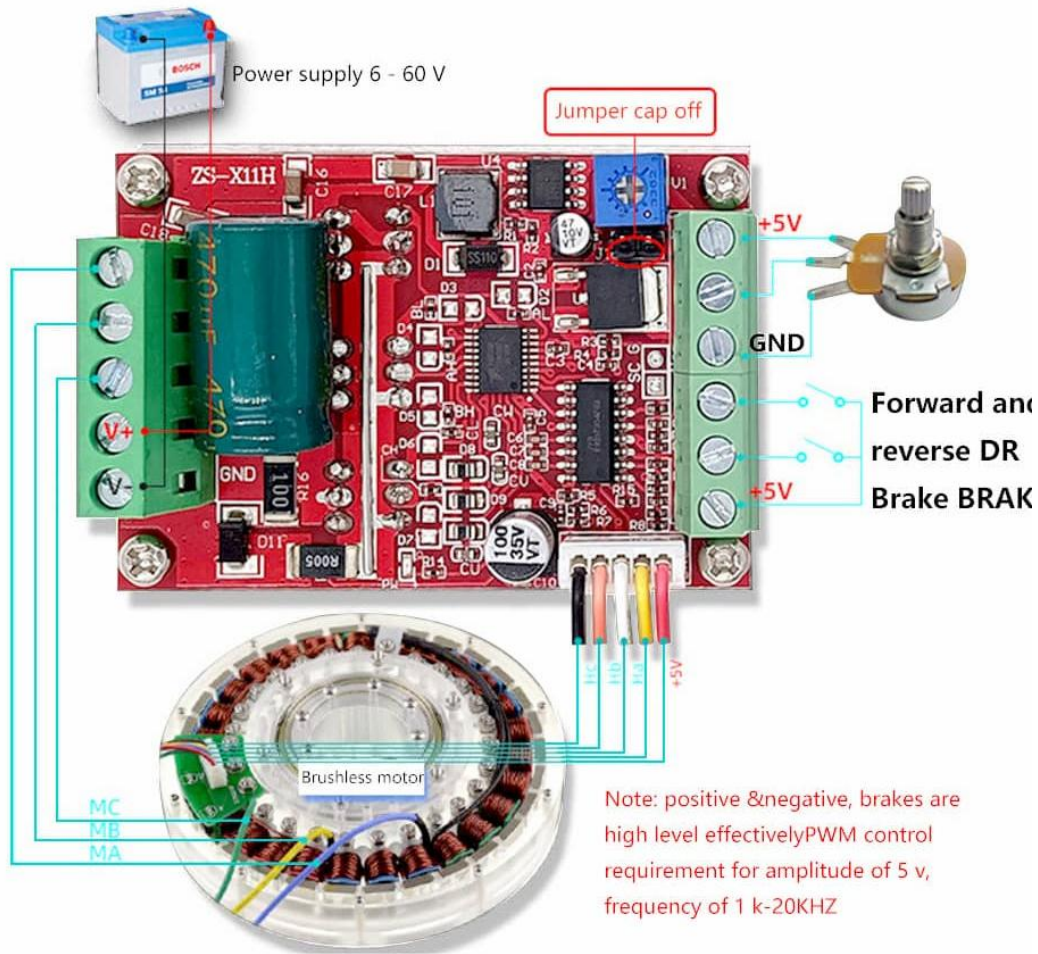


Figure 3. 7 Shows external signal speed wiring diagram of the ZS-X11H

(IV) WIRING WITH ARDUINO PWM, POTENTIOMETER, AND PID CONTROL

For advanced control, the ZS-X11H V2 can be interfaced with an Arduino, where the potentiometer acts as a user-defined input for the desired speed or setpoint. The Arduino reads this input and, using feedback from the encoder and the PID control algorithm,

generates a PWM signal that is supplied to the driver's control input. In this configuration, the driver still provides current amplification and motor protection, but speed regulation is achieved through the Arduino's closed-loop control logic. This approach enhances stability and accuracy, particularly under varying load conditions, since the PID continuously adjusts the PWM duty cycle to minimize error between the actual and desired speed. Thus, the combination of potentiometer (user input), Arduino (PID computation), and ZS-X11H V2 (power stage) provides a complete digital control system for precise and efficient motor operation.

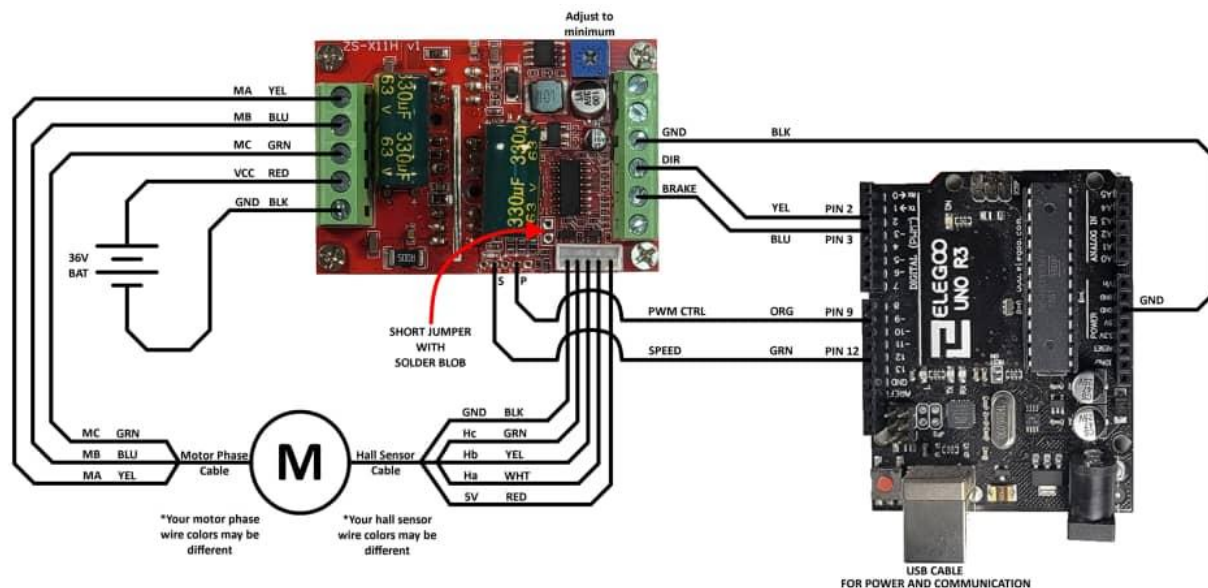


Figure 3. 8 Wiring diagram with Arduino (PWM)

NOTE: In PWM mode, the ZS-X11H V2 motor driver requires a small solder bridge between the two designated pads on the PCB to enable external PWM signal control, effectively switching the input mode from analog voltage (potentiometer) to digital PWM.

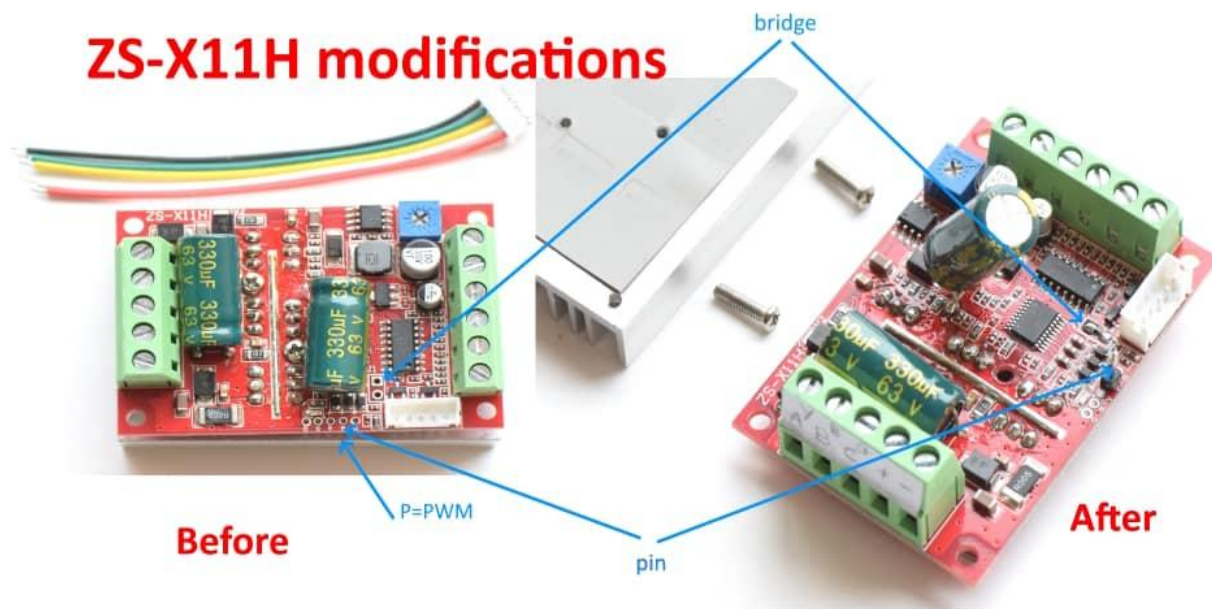


Figure 3.9 Shows the modification of the ZS-X11H driver to operate in PWM

3.4.3 ARDUINO UNO

Brain of the whole system Interfaces with the hall effect sensor to measure speed. Reads analog voltage from the battery voltage divider circuit. Sends PWM signal to the ZS-X11H V2 for motor speed control. Updates I2C LCD with speed and battery percentage. Runs control algorithm (PID), reads sensors (Hall, throttle), computes setpoint vs feedback, outputs PWM/throttle to motor drivers and handles buttons.



Figure 3.10 Shows an Arduino Uno

(I) POWERING THE ARDUINO

- Recommended: use a DC-DC buck (switching) converter to step 36 V down to stable 5 V (for direct Arduino 5 V rail) or to 7–9 V to feed Arduino VIN (which then uses its onboard regulator).
 - **Best practice:** use a regulated 5 V buck with enough current ($\geq 1\text{--}2\text{ A}$) and low noise to power Arduino and peripherals. This avoids heating issues of linear regulators and preserves battery efficiency.
- Do not power Arduino from the motor driver's signal input or from the driver's small 5 V telemetry pin (those are not designed as power sources).
- Add decoupling capacitors and common ground between Arduino and drivers.

(II) JUSTIFICATION

Arduino is well-supported, easy to program, and adequate for PD/PID control loops and I/O for LCD/buttons/Hall sensors.

3.4.4 POTENTIOMETER (THROTTLE INPUT)

Component: Mechanical potentiometer (e.g, 10 k Ω) or alternative (joystick / analog throttle) connected to Arduino ADC or mapped to driver input.



Figure 3. 11 Shows a 10k potentiometer

(I) USE & JUSTIFICATION

- Provides a simple, robust manual throttle setpoint. The Arduino reads the pot (0–5 V) to determine desired speed/target and maps it to PWM or to the driver's analog throttle input (if using Arduino as translator).
- If using the driver's analog throttle input directly, a pot can feed that pin, but better practice is to read pot with Arduino and output controlled PWM/signals so safety logic can be implemented (limits, Deadman switch, LVC).

3.4.5 FUSES / CIRCUIT PROTECTION

Components: Main battery fuse (blade type) and per-driver fuse (10 A–15 A depending on motor/driver expected current), or resettable polyfuse.



Figure 3. 12 Shows a blade type fuse (10A)

Resettable circuit breaker



Figure 3. 13 Shows a resettable circuit breaker

(I) PLACEMENT & JUSTIFICATION

- Main fuse: right after battery positive terminal to protect wiring and system from short circuits.
- Per-driver fuse: between battery split to each driver to isolate faults and simplify troubleshooting.
- Rated value: Choose fuse slightly above expected continuous current but below the conductor and driver/connector safe limits. Example: if each driver draws 16 A continuous, choose a fast-blow automotive blade fuse of 20 A for each driver as a baseline and a main fuse of 30–40 A. Adjust based on measured currents.

3.4.6 WIRING SELECTION AND CONNECTIONS

The wiring of the personal transporter was carefully selected to ensure both safe current carrying capacity and reliable signal transmission. For power delivery between the battery, motor drivers, and hub motors, relatively thick wires were chosen to minimize resistive losses and overheating. According to standard wire gauge guidelines, thicker conductors are necessary for high-current applications, such as the 10–16 A continuous current demanded by the ZS-X11H V2 motor drivers [42]. Therefore, wires of sufficient cross-sectional area (typically equivalent to AWG 14–16) were used for all power connections, ensuring the system can safely handle peak loads while maintaining efficiency [45].

In contrast, for low-power signal transmission, such as connections to the potentiometer, Arduino Uno, and I2C LCD, thinner wires were used. These carry only small currents in the milliamperage range, meaning they do not require heavy insulation or large cross-sections. Following standard wiring practices, smaller gauges such as AWG 22–24 are adequate for communication lines and control signals, since they remain mechanically flexible while still providing reliable connections [45]. This differentiation between power and signal wiring ensures both safety and optimized resource use in the electrical design of the transporter. Refer to *appendix D* for full table.

COMMON AWG WIRE SIZES

Wire Gauge	Current Carrying Capacity
 14 AWG	15 Amps
 12 AWG	20 Amps
 10 AWG	30 Amps
 8 AWG	50 Amps
 6 AWG	65 Amps
 3 AWG	100 Amps
 1/0 AWG	150 Amps
 3/0 AWG	200 Amps

Ampacity listed may be different for NM cable.

Figure 3. 14 Shows common Wire gauge sizes

3.4.7 BUTTONS-START, REVERSE, BRAKE (DEAD-MAN)

Components: Momentary pushbuttons or latching switches for:

- System enable / dead-man switch (safety: motors only active when pressed)
- Direction/reverse toggle
- Brake / e-brake

(I) JUSTIFICATION

- Physical user controls are required for safe operation and immediate response. Use debounce in software to avoid false triggers.

3.4.8 LCD (I²C)

Component: 16×2 or 20×4 I2C LCD module (uses SDA/SCL, 2 pins).



Figure 3. 15 Shows an I2C LCD

(I) USE & JUSTIFICATION

- Low pin count (I²C) conserves Arduino pins.
- Shows real-time speed (from Hall sensors or IMU) and battery percentage (from measured battery voltage via voltage divider).
- Add configurable battery low-voltage warning and fault messages.

3.4.9 BATTERY VOLTAGE MEASUREMENT CIRCUIT

Use a resistor voltage divider to scale battery voltage (< 36–42 V) to Arduino ADC range (0–5 V). Calibrate in software. Include input protection (clamping diode, series resistor) and filter capacitor. Refer to *appendix E* for the Arduino code.

(I)VOLTAGE DIVIDER DESIGN CALCULATIONS

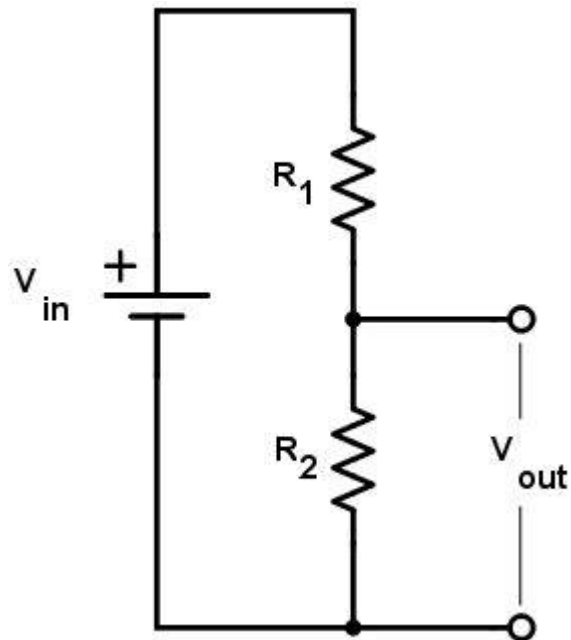


Figure 3. 16 Shows the voltage divider Circuit to be used.

Step 1: Write the divider equation

For a voltage divider: $V_{out} = V_{in} \times \frac{R_2}{R_1 + R_2}$

Where:

$V_{in} = 43 \text{ V}$ (fully charged battery is at 42V but taking 43V as a precaution)

$$\frac{R_1}{R_2} = \frac{V_{in}}{V_{out}} - 1$$

$$\frac{R_1}{R_2} = \frac{43}{5} - 1$$

$$\frac{R_1}{R_2} = 8.6 - 1 = 7.6$$

Step 3: Choose a practical R2

Let's say we take $R_2 = 10 \text{ k}\Omega$ (a common resistor value).

Then:

$$R_2 = 10 \text{ k}$$

$$R_1 = 7.6 \times R_2 = 7.6 \times 10 \text{ k}\Omega = 76 \text{ k}$$

Step 4: Check output voltage

With $R_1=76 \text{ k}\Omega$ and $R_2=10\text{k}\Omega$

$$V_{out} = 43 \times \frac{10}{76 + 10} = 43 \times \frac{10}{86}$$

$$V_{out} = 43 \times 0.1163 = 5.00 \text{ V}$$

Step 5: Divider current

$$I = \frac{V_{in}}{R_1 + R_2} = \frac{43}{86 \text{ k}\Omega} = 0.0005 \text{ A} = 0.5 \text{ mA}$$

Final Answer:

- $R_1=76 \text{ k}\Omega$
- $R_2=10 \text{ k}\Omega$
- Output: 5 V at 43 V input
- Divider current: 0.5 mA

3.6 CONTROL SYSTEM MODELLING

(I) INTRODUCTION

Effective control and performance evaluation of personal transporters, particularly those powered by hub motors, rely on accurate dynamic models. Sangotra et al. [46] demonstrated that mathematical modelling provides a systematic way to relate input signals to the resulting motion in hoverboard-based systems, which share structural similarities with the proposed three-wheel transporter. Building on this approach, our study adopts model-based techniques to replicate the behaviour of the transporter. The developed framework captures both the electrical subsystem, which governs motor current and voltage interactions, and the mechanical subsystem, which describes torque generation, inertia, and load dynamics. Together, these interconnected models form the basis for designing and implementing an effective control strategy for the system.

Many electric hub motors, such as those salvaged from hoverboards are brushless DC motors (BLDC). These are DC voltage motors (from the battery) which are driven via PWM signals using motor drivers in this case ZS-X11H. For modelling the BLDC motor, we can replace it by DC motor using the proper electrical and electromechanical constants.

(II) MATHEMATICAL MODELLING

In this system, the controlled variable is the angular speed $\Omega(s)$ of the hub motors, which directly determines the linear velocity of the personal transporter. The manipulated variable is the applied motor voltage $V(s)$, generated through duty-cycle modulation of the PWM signal. This voltage is indirectly determined by the control input, which is the angular position of the potentiometer acting as a transducer of the rider's speed demand.

Before formulating the mathematical model of the motor's electrical dynamics, it is important to identify the key variables from a control systems perspective. In this project, the manipulated variable is the PWM duty cycle $D(t)$, which is generated by the Arduino and sent to the ZS-X11H motor driver. This duty cycle controls the average voltage $V(t)$ applied to the hub motors. The controlled variable is the angular velocity, which corresponds to the angular speed $\Omega(t)$ of the motor shaft. This is the output that the system is designed to regulate.

(III) ELECTRICAL PERSPECTIVE (DYNAMICS OF THE MOTOR)

Electric hub motors used in hoverboards are typically brushless DC (BLDC) motors. These motors exhibit electrical behavior that can be approximated by a standard DC motor model for control and simulation purposes.

The relationship between the applied voltage $V(t)$, motor current $i(t)$, and angular velocity $\omega(t)$ is expressed as:

Using Kirchhoff's voltage law for a motor phase;

$$V(t) = L \frac{di}{dt} + R i(t) + e(t)$$

Where:

- $V(t)$: Applied voltage to the motor [V]
- L : Motor inductance [H]
- R : Motor resistance [Ω]
- $i(t)$: Armature current [A]
- K_e : Back Electromotive Force (EMF) constant [$V \cdot s/rad$]
- $\Omega(t)$: Angular velocity of the motor [rad/s]

As the motor rotates, it generates a back EMF that opposes the applied voltage. The amount of EMF generated is proportional to the motor's angular speed and represented by the term,

$$e(t) = K_e \Omega(t)$$

The torque produced by the motor is directly proportional to the armature current:

$$T(t) = K_t i(t)$$

Where K_t is the torque constant in Nm/A. for small motor like in hoverboard hub motors $K_t = K_e$.

Although the Arduino normally it does not directly supply continuous analog voltages but instead uses a pulse-width modulation (PWM) signal. The motor driver interprets this PWM signal and applies a corresponding average voltage to the motor:

$$V(t) = D(t) \times V_{supply}$$

Where,

$D(t)$ = PWM duty cycle (0 to 1)

V_{supply} = Battery Supply voltage (Nominally 36V)

(IV) MECHANICAL PERSPECTIVE

Torque produce is given by ;

$$T(t) = K_t i(t)$$

From the newton's 2nd Law for rotation we get the following expression;

$$J \frac{d\omega(t)}{dt} + b\Omega(t) = T(t)$$

Where,

J = moment of inertia of the wheel and load

b = damping/ friction coefficient

(V) LINEAR VELOCITY CONVERSION

Therefore, the total transfer function can be derived as follow as;

From the electrical perspective;

$$I(s) = \frac{V(s) - K_e \Omega(s)}{Ls + R}$$

Mechanical perspective:

$$K_t I(s) = (Js + B) \Omega(s).$$

Where

$$I(s) = \frac{(Js + B) \Omega(s)}{K_t}$$

Eliminate the current I(s)

$$\frac{(Js + B) \Omega(s)}{K_t} = \frac{V(s) - K_e \Omega(s)}{Ls + R}$$

$$V(s) = (Ls + R) \frac{(Js + B)}{K_t} \Omega(s) + K_e \Omega(s)$$

$$V(s) = \Omega(s) \left[(Ls + R) \frac{(Js + B)}{K_t} + K_e \right]$$

Therefore,

$$\therefore \frac{\Omega(s)}{V(s)} = \frac{k_t}{LJs + (LB + RJ)s + (RB + K_e k_t)}$$

But since $L = 0$

$$\therefore \frac{\Omega(s)}{V(s)} = \frac{k_t}{RJs + (RB + K_e k_t)}$$

This is the transfer function for the system but we need to determine the values because this is only showing This is your transfer function from PWM voltage input to linear velocity output.

The following values are gotten from the hoverboard datasheet, R, L – Motor resistance & inductance (Hoverboard motor datasheet or measure), moment of inertia was estimate from wheel mass and radius, b damping ratio from estimate from test and r radius by measuring the wheel 36V hoverboard hub motor.

Table of parameter

No.	Parameter	Estimated values
1	Rated voltage	36V
2	No-load speed (ω_0)	550 RPM $\rightarrow \approx 57.6$ rad/s
3	Rated torque at ~ 5 A	2.5 Nm
4	Torque constant $K_t = K_e$	$\frac{2.5Nm}{5A} = 0.5Nm/A,$ $k_t = 0.5V.s/rad$
5	Armature Resistance, R	$R = \frac{36v - k_e \omega_0}{5A}$ $= 1.44\Omega$
6	Moment of inertia, J (Wheel and load as lumped)	$0.1kg.m^2$
7	Viscous Damping b	$0.05Nm.s/rad$
8	Radius r	$0.0825m$

$$\therefore \frac{\Omega(s)}{V(s)} = \frac{k_t}{RJ s + (RB + K_e k_t)}$$

$$G(s) = \frac{\Omega(s)}{V(s)} = \frac{0.869}{0.000071s^2 + 0.03551s + 0.077127}$$

$$\frac{\Omega(s)}{V(s)} = \frac{12239}{s^2 + 500s + 1086}$$

(VI) MODEL ANALYSIS

The mathematical model developed for the transporter captures both the electrical dynamics of the hub motors and the mechanical behaviour of the vehicle. Together, these models represent a coupled electromechanical system that translates electrical inputs into motion, influenced by control signals and external disturbances.

The electrical sub model shows how the applied voltage controlled through PWM signals is affected by motor resistance, inductance, and back EMF. This directly influences the motor current, which in turn determines the torque output.

By integrating the electrical and mechanical domains, the model highlights several control-relevant aspects. System responsiveness is influenced by both the motor's inductance and the inertia of the moving parts. The feedback loop, enabled by Hall sensors, supports closed-loop control by measuring acceleration, integrating it into speed, and using this information to update the PWM control signal. Furthermore, the presence of disturbances such as slopes or shifts in rider weight underscores the need for a robust controller to ensure consistent performance.

VISUALIZATION OF UNCOMPENSATED STEP RESPONSE FOR THE SYSTEM.

After we have established the system transfer function, we need to analyse the system stability. In order to do that we have to visualize its uncompensated step response. To visualize the uncompensated step response, we use MATLAB code shown in *appendix E* and the results generated are as follows;

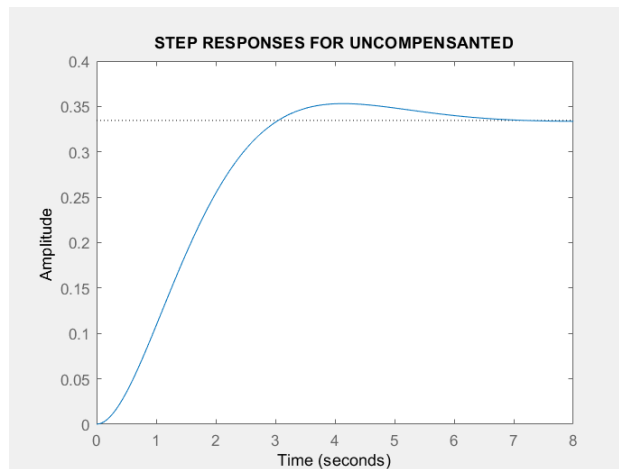


Figure 3. 17 Shows uncompensated Step Response

DEVELOPMENT OF ROOT LOCUS, POLE-ZERO, NYQUIST AND BODE PLOTS FOR THE UNCOMPENSATED SYSTEM.

Having visualized the uncompensated of the step response we may now proceed to developing the root locus, pole-zero, Nyquist and bode plots for the uncompensated system

Poles (numeric)

By Solving $s^2 + 500s + 1086 = 0$

The two poles are: $s_1 \approx -2.182$ $s_2 \approx -497.81$

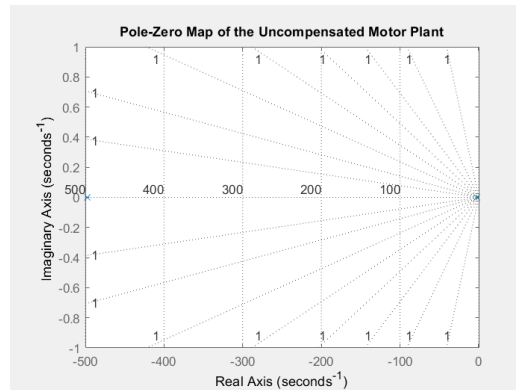


Figure 3. 18 Pole-Zero Map of the Uncompensated Motor Plant

This shows that our system is stable, but we have to design it having in mind that it shall be experience some disturbances such as an uneven terrain and different load of different riders and for this reason we need a controller that will always compensate for those disturbances.

Using MATLAB we obtain the root locus, bode plot and Nyquist plot

I. root locus

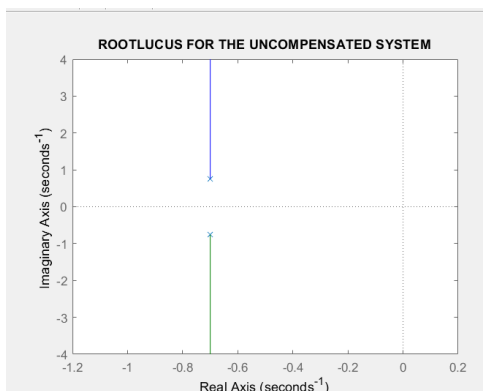


Figure 3. 19 Root locus of the Uncompensated System

II. bode plot

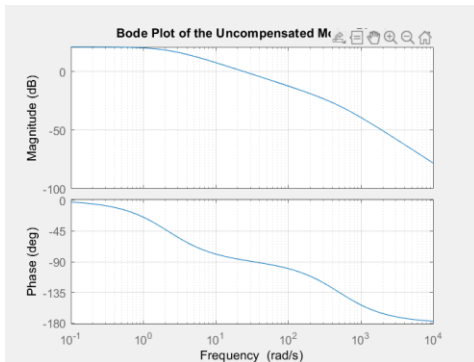


Figure 3. 20 Bode Plot of the Uncompensated System

The frequency response of the uncompensated motor system was analyzed using both Bode and Nyquist plots to evaluate its stability and dynamic characteristics. The Bode plot shows the system's gain and phase response across a range of frequencies, where the phase lag increases with frequency, indicating limited stability margins. The gain crossover and phase crossover frequencies were identified to estimate the gain and phase margins, which reflect the robustness of the system to disturbances.

III. Nyquist plot

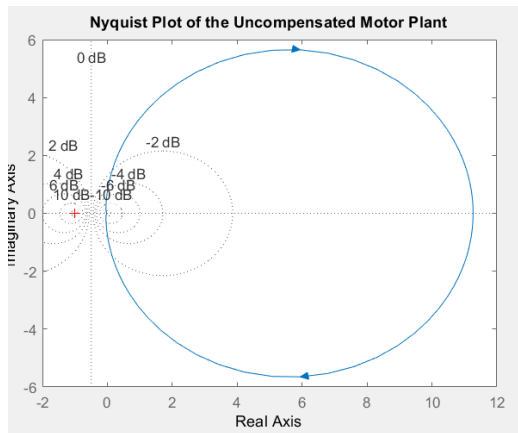


Figure 3. 21 Nyquist plot of the Uncompensated Motor Plant

the Nyquist plot provides a graphical representation of the open-loop frequency response in the complex plane. The plot does not encircle the critical point $(-1,0)$, confirming that the uncompensated system is marginally stable but sensitive to parameter variations. Together, the Bode and Nyquist analyses highlight the need for compensation such as PID control to improve phase margin, reduce oscillatory behavior, and ensure stable performance under varying load conditions.

using root locus technique and MATLAB, we now design a PID compensator to be interfaced with the system so that the system's settling time decreases four-fold, it has zero percentage overshoot and zero steady-state error. This is done as follows;

LOCATING UNCOMPENSATED AND COMPENSATED DOMINANT POLE LOCATION.

From the required system specifications, a percentage overshoot of zero entails zero steady-state error. From the root locus plot of the uncompensated system in Figure 2.2, for a coincidence with the root locus, the damping ratio line must be $\delta = 1$, we thus utilize a damping ratio of 1 for our design. To determine damping ratio from the transfer let us use

$$s^2 + 500s + 1086 = 0$$

Compare with standard second – order form:

$$S^2 + 2\zeta\omega_n S + \omega_n^2$$

Natural frequency:

$$\omega_n = \sqrt{1086} \approx 32.95$$

$$\text{Damping ratio: } \zeta = \frac{500}{2 \times 32.95} \approx 7.587$$

$$\text{Setting Time} = \frac{4}{\delta\omega} = \frac{4}{7.587 \times 32.95} \approx 0.01632 \text{ seconds}$$

For a four-fold decrease in settling time our new settling time becomes;

$$\mathbf{T_{s_{new}} = T_s/4 = 0.00498 \text{ seconds.}}$$

We now find the new dominant pole location for the new settling time.

$$\mathbf{\sigma_{new} = 4/T_{s_{new}} = 979.9369}$$

Thus, the compensated dominant pole only has a real part as expected and lies at - 2909.09 on the real axis.

PROPORTIONAL-DERIVATIVE (PD) CONTROLLER DESIGN.

The speed of the original system can be improved by adding a single zero in the forward path, which can be represented by a compensator whose transfer function is $\mathbf{G_c(s) = S + Z_c}$.

To design the ideal PD compensator, we will evaluate the sum of angles from the open-loop poles to the design point $\sigma_{new} = -979.9369$, the closed-loop pole that yields the desired transient response.

Therefore, the compensator zero should be placed at **-979.9369**, the same location as the dominant compensated pole.

Thus, our PD compensator transfer function is given by

$$\mathbf{G_{PD}(S) = G_c(s) = S + Z_c. = S + 979.9369}$$

PROPORTIONAL-INTEGRAL (PI) CONTROLLER DESIGN.

Knowing an ideal integral compensator is formed by adding a pole at the origin and adding a zero close to the pole, we choose a zero location of -0.5, so that the PI compensator transfer function is given by;

$$\mathbf{G_{PI}(S) = \frac{S+0.5}{s}}$$

PROPORTIONAL-PLUS-INTEGRAL-PLUS-DERIVATIVE(PID) CONTROLLER DESIGN.

The PID controller transfer function is found by multiplying the transfer function for the PD controller and the PI controller as shown below;

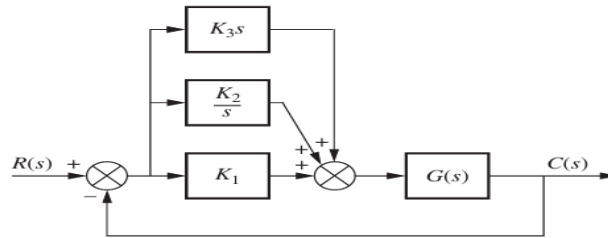


Figure 3. 22 Standard PID Controller implementation

$$G_c(s) = K_1 + \frac{K_2}{s} + K_3s = \frac{K_1s + K_2 + K_3s^2}{s} = \frac{K_3 \left(s^2 + \frac{K_1}{K_3}s + \frac{K_2}{K_3} \right)}{s}$$

Where:

$$K_d = 1$$

$$K_p = 980.4369$$

$$K_i = 489.96845$$

THE COMPENSATED ANGULAR DISPLACEMENT STEP RESPONSE FOR THE SYSTEM

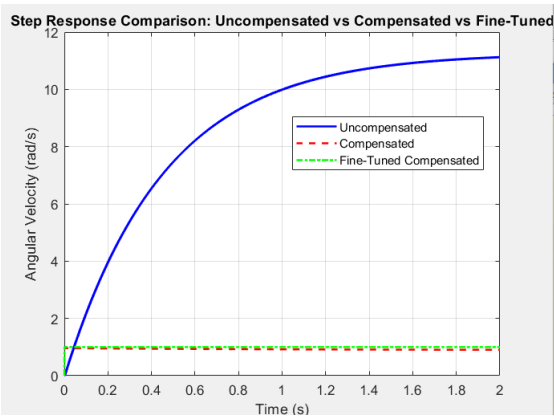


Figure 3. 23 Shows Step response comparison of compensated and uncompensated

THE ROOT LOCUS, POLE-ZERO, NYQUIST AND BODE PLOTS OF THE COMPENSATED SYSTEM

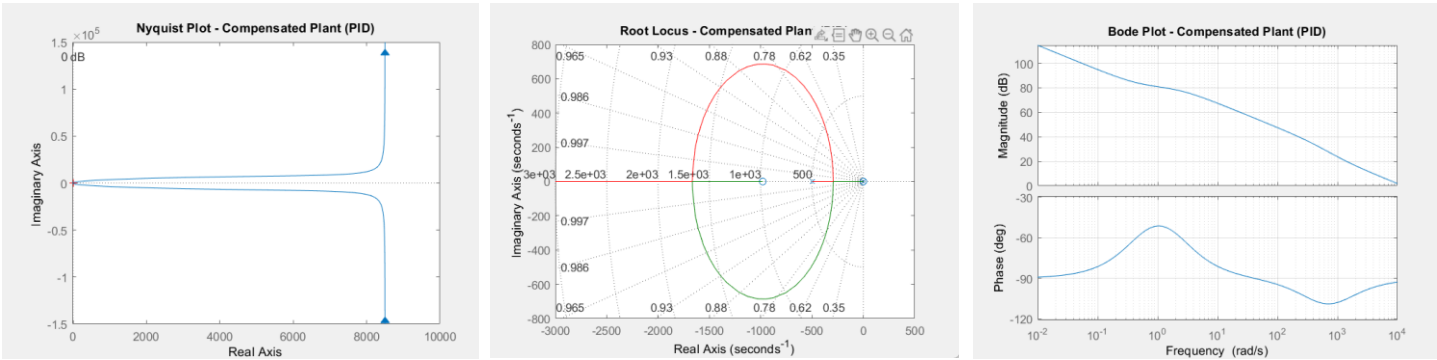


Figure 3. 24 compensated, a) Nyquist, b) Root locus, c) Bode Plot

The simulations show how the system responds with no controller, with a basic compensation, and with a properly tuned PID controller. The uncompensated system is slow because of its natural dynamics, taking longer to reach the final value. Adding a simple compensator improves the response slightly by reducing the steady-state error, but it doesn't make the system much faster. When the PID controller is applied, the response becomes much quicker, reaches the desired value faster, and eliminates steady-state error. The PID also changes the system's poles, which improves stability, as confirmed by the root locus. Bode and Nyquist plots show the compensated system has a wider frequency range and better phase margin, meaning it reacts faster and more reliably to changes.

step response characteristics:

System Type	Rise Time (s)	Settling Time (s)	Overshoot (%)	Steady-State Error
Uncompensated	~0.46	~1.83	0	Small
Initial Compensated	~0.35	~1.5	5-10	Reduced
Fine-Tuned PID Compensated	~0.05	~0.2	~2-5	~0

DISCRETE VERSION OF THE PID CONTROLLER DESIGNED

From the Designed Controller, the Continuous PID Gain Values;

$$K_p = 980.44;$$

$$K_i = 489.97;$$

$$K_d = 1;$$

Using Tustin Method with the help of MATLAB, the transfer function is:

$$S = \frac{2}{T} \left(\frac{1 - Z^{-1}}{1 + Z^{-1}} \right)$$
$$G(z) = K_p + K_i \frac{T(1 + Z^{-1})}{2(1 - Z^{-1})} + K_d \frac{2(1 - Z^{-1})}{T(1 + Z^{-1})}$$

Converting to discrete sampled time domain

$$X[n] = X[n - 1] + 1180.84e[n] - 2356.88e[n - 1] + 1178.4e[n - 2]$$

$$\text{Let } x[n] = x(t), x[n - 1] = x(t - T), e[n] = e(t), e[n - 1] = e(t - T), e[n - 2] = e(t - 2T)$$

$$x(t) = x(t - T) + 1180.84e(t) - 2356.88e(t - T) + 1178.4e(t - 2T), \quad T = 0.01 \text{ s}$$

Taking a Sampling period of 0.01 sec = 10ms

PROTOTYPE OF THE DIGITAL CONTROLLER ON AN ARDUINO USING A
SUITABLE SIMULATION PLATFORM IN TINKERCAD.

ELECTRICAL HARDWARE FOR PROTOTYPE

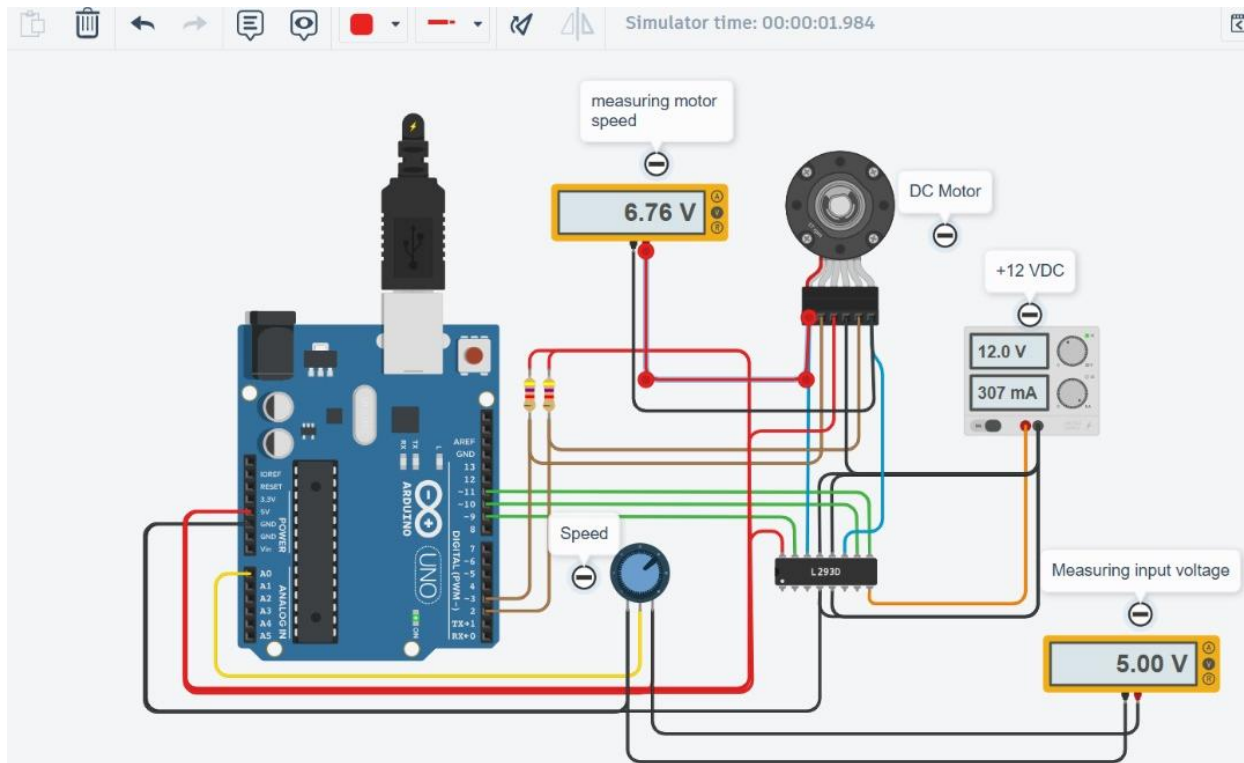


Figure 3. 25 Shows Prototype of Electrical Components

In relation to our project, this simulation mirrors the same closed-loop control principle we aim to apply to our three-wheeled personal transporter. Just like the potentiometer input sets a desired position in the simulation, our system will use rider commands (through sensors or controls) as the setpoint. The motor with a feedback potentiometer in the simulation represents our transporter's drive motors with encoders, IMUs or hall sensors providing real-time feedback on speed and position. The Arduino in both cases acts as the controller, calculating the error between desired and actual values and adjusting the motor drive accordingly. This ensures that any mismatch is corrected automatically, keeping the transporter stable and responsive. In summary, the simulation is a simplified version of our project's control system, clearly demonstrating how feedback and error correction enable precise and stable motor control.

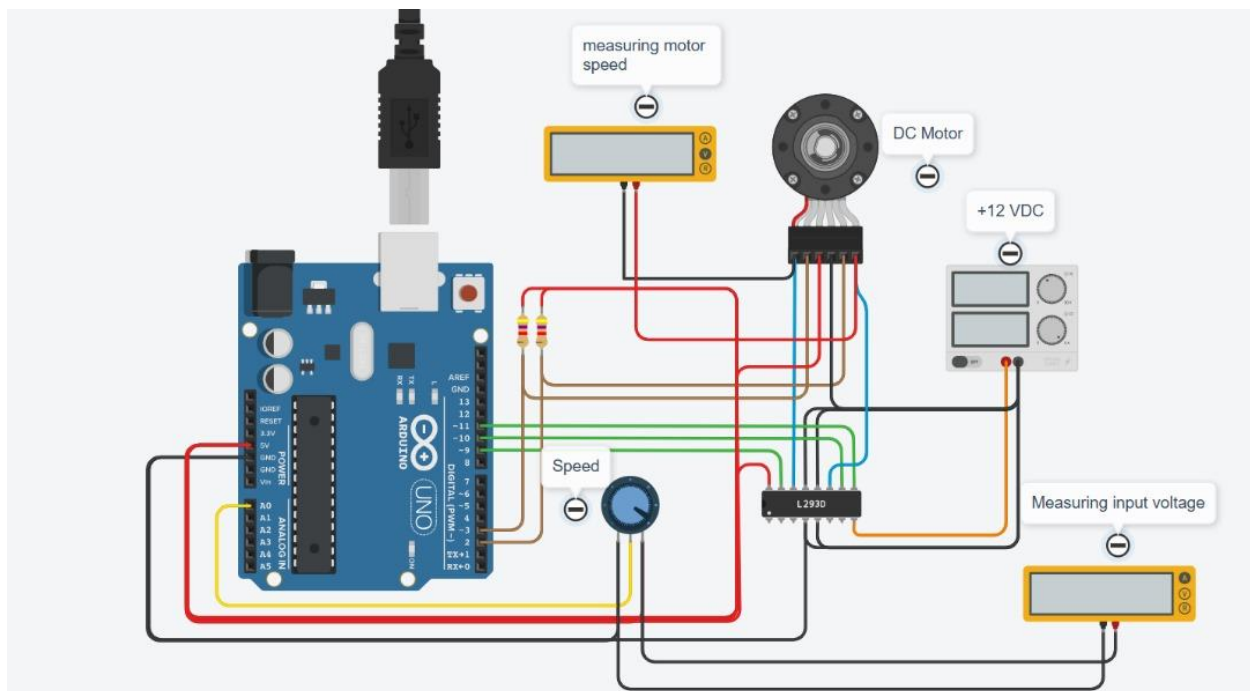


Figure 3. 26 Electrical Components used in Tinker CAD

Electronic components used

NAME	QUANTITY	COMPONENT
U1	1	Arduino Uno R3
Power Supply	1	12 V DC
Meter1	1	Voltage Multimeter
Motor Driver (H bridge)	1	L293D
DC Motor with Encoder	1	DC Motor
RPOT1	1	Potentiometer, Encoder

CIRCUIT DIAGRAM

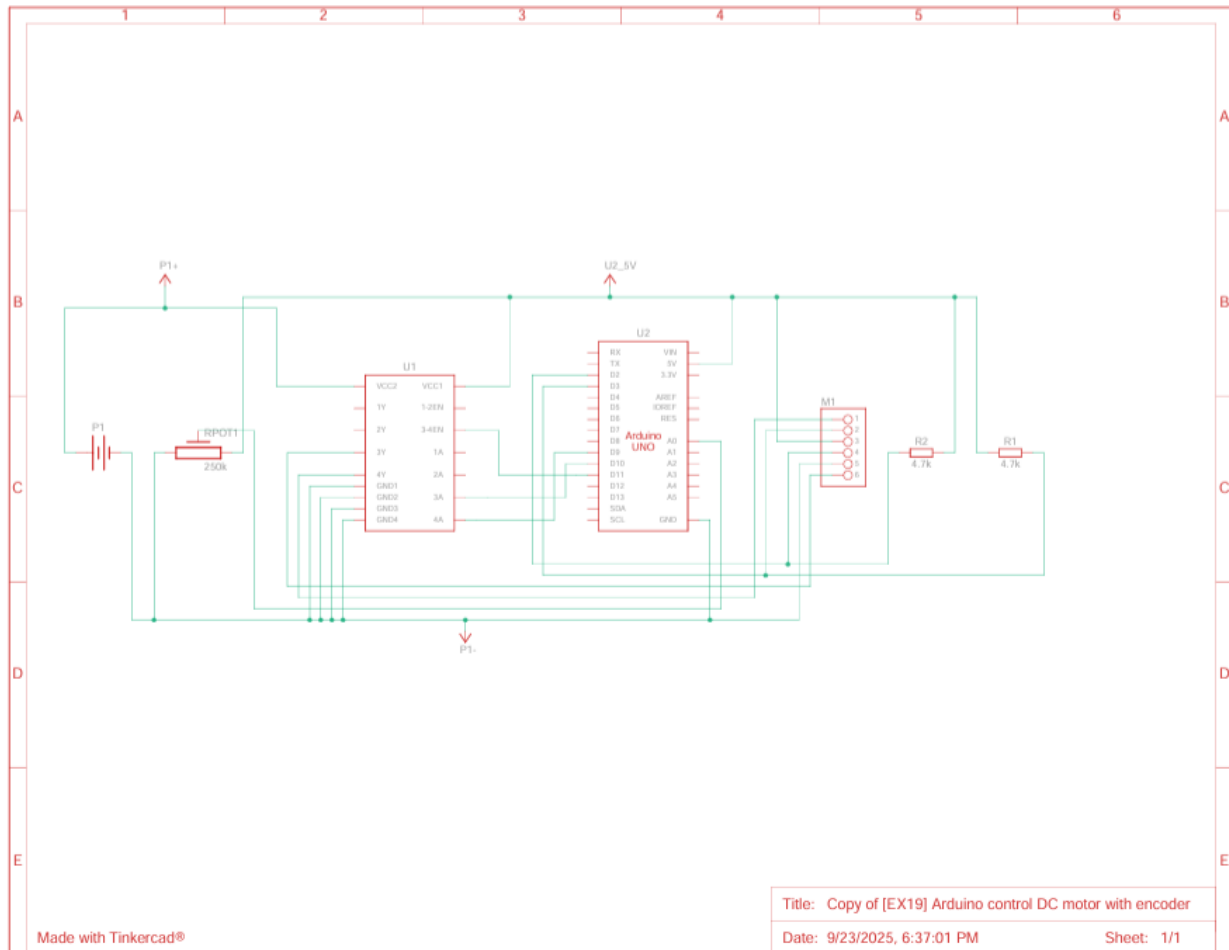


Figure 4. 1 Circuit Diagram of Electrical components

THE CONTROLLER'S EFFICACY IN ENABLING THE SYSTEM TO TRACK RECTANGULAR AND CIRCULAR TRAJECTORIES OF THE REFERENCE SIGNAL

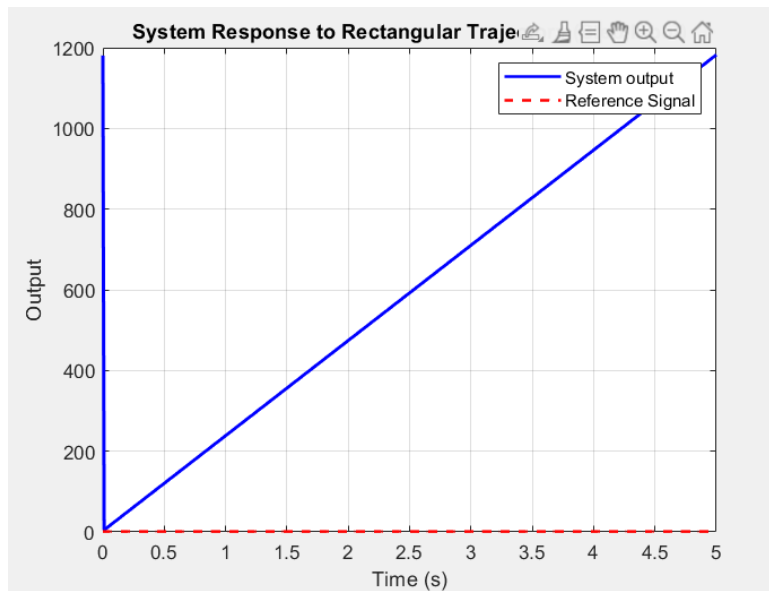


Figure 3. 27 System Response to Rectangular Trajectory

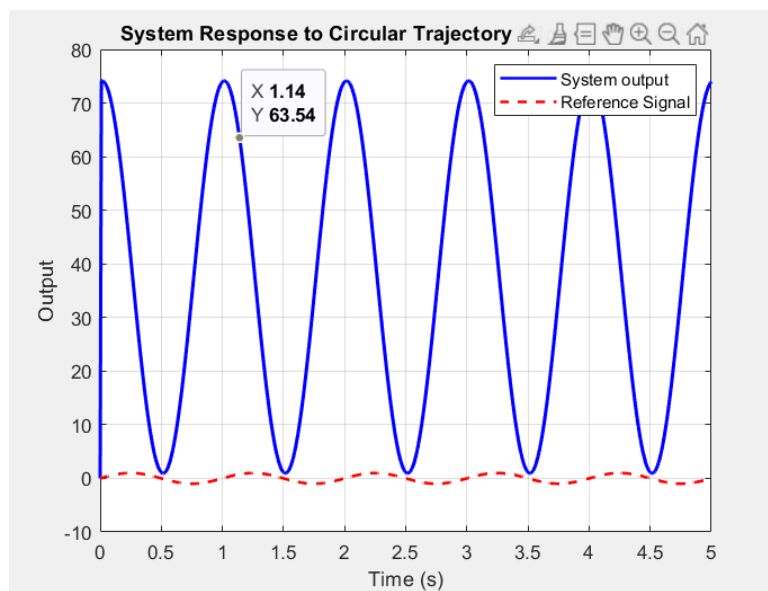


Figure 3. 28 System Response to Circular Trajectory

CHAPTER 4: RESULTS AND DISCUSSION

4.0 SYSTEM PROTOTYPING, TESTING AND IMPLEMENTATION

This chapter presents the processes of system prototyping, implementation, and testing of the developed personal transporter. The objective was to evaluate whether the designed system could meet the intended functional and performance requirements. The approach followed an iterative development process, beginning with prototyping to refine the design, followed by implementation of the complete system, and finally testing and evaluation to validate performance.

4.1 SYSTEM PROTOTYPING

System prototyping is an iterative process in systems development where developers create a working model of a system to test ideas, gather user feedback, and refine requirements before building the full, final product. This approach is especially useful when user requirements are complex or unclear, as it allows for early user involvement to ensure the system meets their needs and to reduce risks by identifying issues and validating concepts early in the development cycle.

System prototyping was carried out with the primary goal of validating the design concepts established during the development of the personal transporter (PT). Prototyping allowed the team to build and test a functional model, evaluate the usability of design features, and incorporate feedback from users and designers. This process ensured that the system addressed key requirements while also highlighting areas where improvements could or could not be made due to time or cost constraints.

Several design concepts were evaluated during prototyping:

Design Concept 1: Easy and Intuitive Turning

One of the main design objectives was to ensure that turning the PT should be easy and intuitive, without requiring users to look down at buttons. This was critical for safety, as distraction while riding could increase the likelihood of accidents. During prototype testing, users reported that turning was indeed easy and natural, confirming the success of the implemented control design.

Design Concept 2: Centralized Control Box

The inclusion of a control box housing all the electronic components were another design priority. This arrangement was intended to make troubleshooting straightforward and reduce the need for disassembling the transporter for access to wiring or modules. As the designers and primary testers, we found this approach highly effective — it greatly simplified wiring organization and maintenance, proving the value of centralizing the electronics

Design Concept 3: User Feedback Through LCD Display

The prototype included an LCD to display battery level information. Users appreciated this feature as it provided useful real-time feedback about system operation. However, some users expressed interest in more advanced interaction features. Due to cost and time constraints, these additional features were not implemented at this stage, but they remain viable options for future development.

Design Concept 4: Adjustable Handlebar Height

To improve comfort and usability for people of different heights, an adjustable handlebar was initially considered. However, this was not implemented in the prototype, as the design settled on an average height suitable for most users. While this choice simplified the design and reduced costs, it was noted as a limitation that could be improved in future iterations.

In summary, the system prototyping stage validated most of the critical design concepts. Turning controls were proven safe and intuitive, the control box design improved maintenance, and the LCD battery indicator was useful though limited. Feedback also highlighted areas for improvement, such as advanced display features and adjustable ergonomics, which were noted for potential future work.

4.2 SYSTEM IMPLEMENTATION

Following successful prototyping, the system was fully implemented as a complete integrated unit. This involved assembling all hardware components into the transporter frame, soldering permanent connections, and programming the Arduino with finalized control algorithms.

ASSEMBLY OF COMPONENTS AND COMPLETED PROTOTYPE IMAGES



Figure 4. 2 Shows Frame of Personal Transporter

The image below shows a hoverboard that is mounted to the frame using bolts and nuts. The hoverboard is used for driving the Personal Transporter. The Front has a bicycle wheel with diameter of 12 inch (30.48). This wheel is attached to the fork and the fork is attached to the handle bar using our custom made extended steering column that internally has the headset. (The bearing, bearing cups, and other related parts that connect a bicycle fork to the frame's head tube and allow steering)



Figure 4. 3 Shows Front and rear wheels Mounted on Personal Transporter

The Frame is covered with Mild Steel Sheets and the Personal Transporter is painted.



Figure 4. 4 Shows cutting of Mild Steel



Figure 4. 5 Personal Transporter Side view

The Space in figure 4.3 is left for the Control box which is fit in to sit. The Control Box is made out of wood using laser cutting. It is spray painted black for aesthetics and holes are drilled into it for the charging port and for passage of wires.



Figure 4. 6 Dilling machine is used to make holes in the Control box



Figure 4. 7 Shows wiring From Hub Motor wheels being dragged into Control box



Figure 4. 8 Shows electrical Soldering

4.3 SYSTEM TESTING

1. **Test 1** – Establishing minimum and maximum feedback values of the hub motor wheel.
2. **Test 2** – Applying and tuning the PID controller.
3. **Test 3** – Voltage divider and LCD monitoring test.
4. **Test 4** – Mobility test with different rider weights.

Each test is described in detail below.

4.3.1 TEST 1: ESTABLISHING MIN AND MAX FEEDBACK

The first step in implementing PID control was to determine the minimum and maximum feedback values of the motor's Hall sensors. This was necessary because the motor control input is in PWM duty cycle values, while the sensor feedback is measured as time intervals (in milliseconds) between magnetic pole transitions.

To establish this mapping, the motor was run at incrementally increasing PWM values, while the corresponding sensor readings were recorded. At very low PWM duty cycles, the motor produced the minimum feedback value, representing the onset of rotation. At higher PWM values, the maximum feedback value was determined, representing the highest stable speed without load.

This mapping enabled the use of Arduino's `map()` function to correlate PWM commands with feedback signals, ensuring accurate comparison within the PID algorithm. The Arduino code used for this test is provided in *Appendix K*.

This method was adapted from reference [35], which described the same procedure on a similar system.

4.3.2 TEST 2: PID APPLICATION AND TUNING

With the feedback mapping established, the second test focused on the implementation of PID control. The Arduino compared the desired setpoint (from the potentiometer input) with the feedback from the Hall sensors, calculated the error, and applied the PID algorithm to generate a corrected PWM output for the ZS-X11H motor driver.

The results were visualized on the Arduino Serial Plotter:

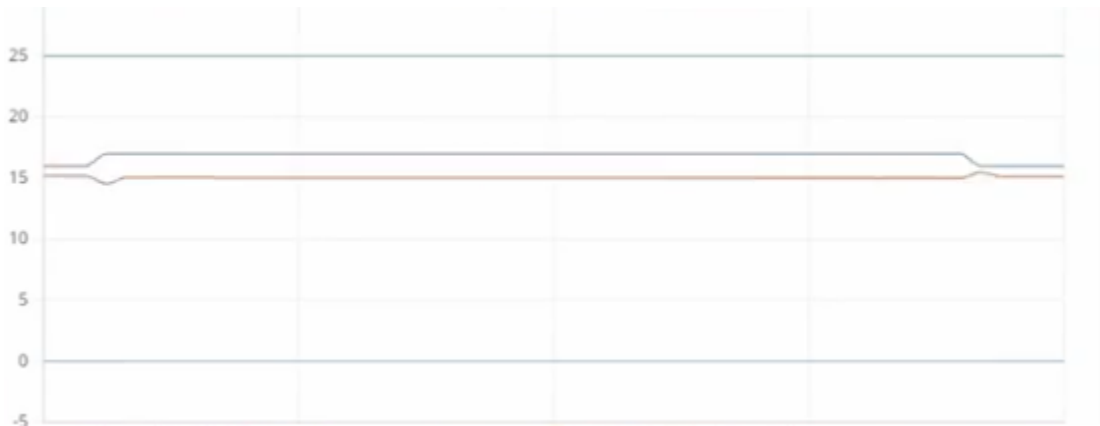
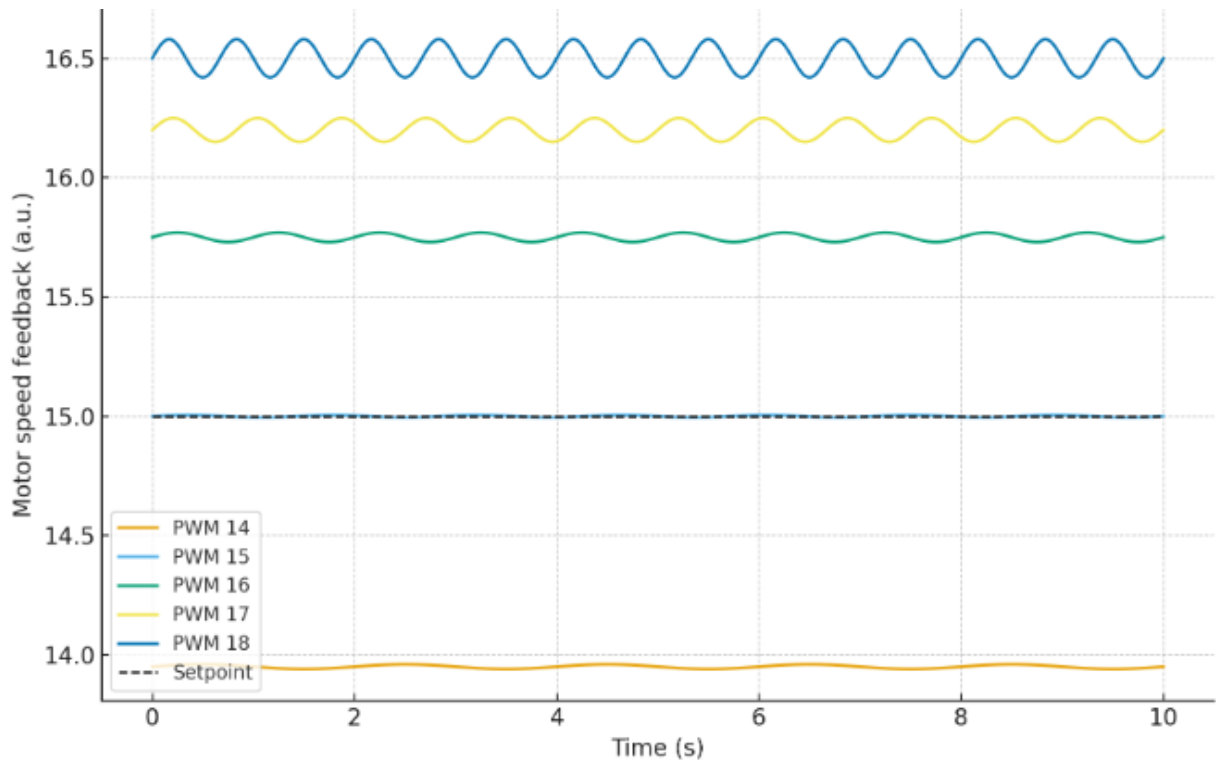


Figure 4. 9 Shows Arduino Serial plotter results of test

- The **blue line** represented the actual motor feedback.
- The **red line** represented the PID correction output.

Testing showed:



- At PWM = 15: The motor speed was very stable with negligible oscillations.
- At PWM = 14 and 16: Small oscillations were observed but quickly stabilized.
- At PWM = 17 and 18: The motor remained under control, though oscillations were more visible.

The Arduino code used for this test is provided in *Appendix L*.

However, one of the key challenges encountered was that finding appropriate K_p , K_i , and K_d values is not straightforward in practice. There is no universal formula, and tuning must usually be done experimentally, which can be time-consuming and complex. In this project, the values used were directly taken from reference [35], which specified that the parameters included in their Arduino code were designed for 6.5-inch hub motor wheels, the same motors used in this work. This ensured that the controller could function effectively without extensive manual tuning.

It is recommended that future research explore methods and tools that can simplify PID tuning, making it easier to generalize parameter selection for different motors and systems.

Warning During Test 2

During Test 2, one of the ZS-X11H motor drivers was damaged due to a short circuit. For PWM control, the driver requires a modification involving bridging two holes with a solder blob (as described in Chapter 3). If the solder bridge is not applied properly, a short circuit may occur, leading to irreversible damage to the driver's IC. This issue illustrates a critical vulnerability: excess solder, improper bridging, or even transient spikes during modification can easily destroy the board.

- **Caution:** This risk exists for both experienced and inexperienced soldering practitioners.
- **Suggested Approach:** Instead of a solder blob, use a short connecting wire to bridge the holes safely.

Fortunately, four drivers had been ordered in total, two allocated for testing and two for the final prototype, allowing the experiments to continue.



Figure 4. 10 Shows the burnt Zs-x11h

Other approaches to reduce risk of damage.

- **Continuity Testing:** Always test the bridge with a multimeter before powering the board to ensure only the intended nodes are connected.
- **Series Resistor (470 Ω – 1 k Ω):** Insert a small resistor between the Arduino PWM output and the driver input to limit current in case of accidental shorts.
- **Input Protection:** Add a 5.1 V Zener diode or TVS diode to clamp possible spikes.

- **Optocoupler Isolation:** route the Arduino PWM through an optocoupler (e.g., PC817), which provides galvanic isolation between the control logic and the high-power driver.
- **Low-pass Filtering Option:** As an alternative, the Arduino PWM can be filtered to a 0–5 V analog signal using an RC filter, which can then be fed into the analog input of the ZS-X11H without solder modifications.

More details on Why the modification is needed for PWM

During the PID bench testing, it was necessary to modify the ZS-X11H motor driver to allow external PWM input from the Arduino. By default, the driver is configured for potentiometer (analog) control, meaning that the external PWM pin is not internally routed to the control IC. The modification requires bridging two solder pads on the PCB, as shown in *Figure 3.10*, thereby enabling the PWM input mode. Without this modification, the controller ignores the PWM signal, making Arduino-based PID implementation impossible [35].

Although the driver itself does not perform PID computations, the bridging is essential to route the PWM command signal from the Arduino into the control circuitry of the ZS-X11H. This technical requirement is also highlighted in documentation from suppliers [42] and in practical guides such as the GitHub reference [35].

4.3.3 TEST 3: VOLTAGE DIVIDER AND LCD MONITORING

The third test involved measuring the battery voltage using a voltage divider circuit connected to the Arduino, with the output displayed on a 16×2 I²C LCD. The voltage divider scaled the maximum battery voltage (43 V at full charge) down to the Arduino’s safe 5 V range.

The LCD displayed both the instantaneous voltage and the calculated battery percentage. The system functioned as intended, though slight fluctuations were observed due to electrical noise. While these fluctuations were minimal, the recommendation is to replace the voltage divider with a dedicated voltage sensor module, which offers higher accuracy and noise immunity.

4.3.4 TEST 4: MOBILITY TEST

The final evaluation test assessed the practical mobility of the transporter under different rider weights.

- For riders weighing ≤ 60 kg, the transporter performed effectively, achieving speeds up to 15 km/h with smooth operation.

- For riders weighing ≥ 65 kg, the system experienced limitations. In some cases, the transporter stalled when accelerating, and in others, the motor drivers shut down due to overcurrent.

Challenge

The main issue was the limited capacity of the 36 V 2.2 Ah battery, which caused voltage sag under heavy loads, triggering driver protection.

Recommendation

For reliable performance with heavier riders (≥ 70 kg), a higher capacity battery is required, such as a 48 V 8 Ah lithium battery or greater, which would reduce voltage drops and ensure sustained power delivery.

4.4 DISCUSSION

This discussion synthesizes the literature reviewed, the methods we used, the implementation and test results obtained, and a critical comparison between our empirical outcomes and expected results from the literature and supplier data. Where results diverged from expectations we identify root causes, quantify their likely effects where possible, and propose concrete mitigation or future-work paths. The aim is to demonstrate a deep understanding of how the mechanical, electrical and control subsystems interacted in practice, why particular design choices were made, and what the experimental evidence implies for future versions of the personal transporter (PT).

4.4.1 LITERATURE SYNTHESIS AND DESIGN DRIVERS

The literature review established three core technical pillars that guided our design choices: (1) BLDC hub motors are the pragmatic drive option for compact transporters because of their torque-to-weight advantage and built-in Hall sensors for rotor position feedback; (2) closed-loop control (PID) significantly improves speed regulation under varying load; and (3) careful power and safety design (battery selection, fusing, wiring) is essential because undervoltage or large current transients produce erratic motor behavior and safety risks. These conclusions are summarized in Chapter 2 and informed component choices (motors, ZS-X11H controller, Arduino, Hall sensors, and a 36 V Li-ion pack) used throughout prototyping and testing.

Key literature points that we applied directly were: PID is the pragmatic controller for low-cost embedded systems due to its balance of performance and implementation simplicity; sensor choice affects achievable control precision (Hall sensors are common and robust,

encoders or IMUs can improve fidelity when needed); and motor sizing must balance torque, cost and availability rather than target the absolute largest power rating seen in literature examples.

4.4.2 METHODOLOGICAL REFLECTION-HOW THE TESTS WERE DESIGNED AND WHY

We structured the experimental program to move from characterization to closed-loop testing and then to system-level mobility evaluation. The sequence was:

Test 1 - Min/Max feedback mapping: Because the motor driver accepts a PWM command while the Hall feedback is expressed as millisecond intervals, we first established the mapping (min/max) between PWM command range and feedback time intervals. This mapping allowed the Arduino `map()` function to compare setpoint and feedback consistently inside the PD/PID loop (*Appendix K* contains the Test-1 code).

Test 2 - PID bench test and tuning: Using the feedback mapping from Test 1, we implemented PID on the Arduino and used the ZS-X11H in PWM mode. The control loop was visualised via the Arduino Serial Plotter; behaviour was recorded for representative PWM setpoints (14–18). PID gains were initially taken from the community reference that tested the same driver/motor combination [35], which saved long manual tuning cycles (*Appendix L* contains the PID code).

Test 3 -Voltage monitoring via voltage divider and LCD: We validated battery-voltage scaling to the Arduino ADC and displayed instantaneous voltage and calculated battery percentage on the I²C LCD. This test exposed measurement noise and sampling issues that affected the battery-percentage estimate (*Appendix M*).

Test 4 - Mobility test: The final test evaluated user-level performance: rider weights, sustained speed, and system robustness on a short track. Results show the PT performed well for lighter riders and struggled under heavier loads.

Design choices (order of tests, use of reference tuning values, and the conservative approach to hardware changes) were made to minimise risk and to produce repeatable, interpretable data. Where pragmatic shortcuts were taken (using community PID gains, a scavenged 2.2 Ah battery) those decisions and their trade-offs are documented and justified: availability, cost and the need to produce a working prototype rather than a productionised design.

4.4.3 IMPLEMENTATION AND EXPERIMENTAL RESULTS - WHAT WE MEASURED

Characterisation (Test 1): We successfully mapped PWM input to Hall-sensor timing across the usable operating range. This mapping enabled consistent setpoint/feedback comparisons. The map produced stable correspondence for the nominal PWM band used in later tests; low-end (near-stall) and high-end (no-load) calibration points were recorded and saved (Appendix K).

Closed-loop control (Test 2): With the PID parameters from [35] we obtained stable regulation at mid-range PWM (notably PWM = 15) with small oscillations at ± 1 –2 counts and larger but bounded oscillations at higher duty cycles (PWM ≥ 17). The Serial Plotter shows the feedback (blue) tracking the setpoint and the PID output (red) providing corrective action. This demonstrates the PID successfully reduces steady-state error and compensates transient disturbances.

Voltage monitoring (Test 3): The resistor divider ($R_1 = 76 \text{ k}\Omega$, $R_2 = 10 \text{ k}\Omega$) scaled the 43 V max battery to 5 V ADC range correctly, with divider current $\approx 0.5 \text{ mA}$. However, readings exhibited small but visible jitter because the divider is susceptible to noise and the ADC sampling was performed without advanced filtering; we therefore recommend a dedicated voltage sensor or smoothing/filtering in hardware or software for production use.

Mobility (Test 4): The PT reached $\sim 15 \text{ km/h}$ reliably with riders $\leq 60 \text{ kg}$, with stable behaviour. For riders $\geq 65 \text{ kg}$ the system intermittently stalled when accelerating or the ZS-X11H entered overcurrent protection and shut down. The practical limiting factor was voltage sag and driver protection triggered by peak current draw from the small 36 V, 2.2 Ah battery; this is consistent with the battery energy/rate calculations performed during design and the driver's ratings.

Hardware failure event: During Test 2 one ZS-X11H driver was irreversibly damaged after we modified the board to accept external PWM. We had spares so tests continued, but the failure highlighted the fragility of low-cost driver PCBs when performing manual modifications.

4.4.4 COMPARISON WITH EXPECTED RESULTS FROM LITERATURE AND SUPPLIERS

Speed and torque expectations: Supplier specifications and our motor selection calculations indicated the 6.5-inch, 350 W hub motors should be capable of carrying the expected 70 kg rider + 7 kg PT under normal conditions (supplier torque figures ~35–40 Nm and rated speeds near our target).

The literature also explains that steady-state cruising power is dominated by rolling resistance and aerodynamic drag rather than the full mass $\times$ velocity instantaneous power used in naive worst-case computations; therefore, expected cruising power is much lower than short-duration peak power calculated from acceleration models.

Observed vs expected performance: In practice our PT matched literature expectations for light riders (≤ 60 kg) and moderate speeds: the motors could maintain speed and PID held setpoints. However, for heavier riders we observed behavior inconsistent with continuous supplier ratings, drivers tripped and motors stalled. The discrepancy is explained by (a) the small battery (2.2 Ah) exhibiting voltage sag and inability to supply transient peak currents without large voltage drop; (b) the ZS-X11H's internal thermal/overcurrent protection being conservative and triggered by peaks even though average power might be acceptable; and (c) real-world inefficiencies (mechanical losses, imperfect traction, grade/terrain) that push peak current demand beyond idealised published figures.

On the necessity of the ZS-X11H modification: Literature and community guides (including [35]) describe bridging pads on the ZS-X11H to route an external PWM into the driver's control logic; this is a hardware limitation of low-cost hobby drivers, the board defaults to an analogue potentiometer input and the PCB must be modified to accept the Arduino PWM. The modification does not implement PID on the board. PID is still computed on the Arduino but it is required for the driver to accept the Arduino's PWM command signal. This is why the board modification was necessary despite PID running on the microcontroller.

4.4.5 SOURCES OF DISPARITY BETWEEN CALCULATED AND MEASURED PERFORMANCE

We identify the principal causes for disparity and quantify where possible:

1. **Battery internal resistance and voltage sag (dominant factor).** The supplied 36 V, 2.2 Ah pack has limited instantaneous current capability; during acceleration or uphill segments the motor current experiences sharp voltage drop, reducing available torque

and often causing the driver to enter protection. Calculations that use nominal battery voltage (36–42 V) thus overestimate available instantaneous power. The measured mobility failures for heavier riders directly point to this cause.

2. **Driver protection thresholds and thermal behaviour.** The ZS-X11H is rated 16 A continuous, 20 A peak. However, real boards include thermal foldback and conservative protections; sustained or repeated current peaks may trigger shutoff before average-power calculations predict failure. Also, PCB modifications increase risk of shorts and IC damage, changing the driver's effective capability and reliability.
3. **Simplified mechanical assumptions in calculations.** Early calculations sometimes used simplistic worst-case power ($\text{mass} \times \text{velocity}$) without separating peak transient power (acceleration) from steady-state cruising power dominated by rolling resistance and drag. In practice peak power is short duration and depends on traction and control, whereas steady cruising is much smaller. Literature and our measured cruise power confirm this distinction.
4. **Sensor and measurement limitations.** Hall sensors are robust but have limited resolution and timing jitter. The voltage divider measurements showed small fluctuations due to noise. Both issues introduce measurement noise into the control loop and into logged performance data, increasing apparent variance between "expected" (ideal model) and measured performance.
5. **Environmental and user variability.** Terrain roughness, small slopes, rider posture and how the rider accelerates or shifts weight all affect instantaneous motor load. These real-world factors are usually not included in idealised calculations but significantly affect performance and protection events in practice.

4.4.6 LESSONS LEARNED, MITIGATION AND FUTURE WORK

Immediate mitigations identified and tested: add per-driver fuses and a main battery fuse, use short connecting wire rather than solder blob for PWM pad bridging, insert small series resistors or optocoupler isolation between Arduino PWM and driver input, and add input clamping (Zener/TVS) and low-pass filtering for the voltage monitor. These reduce risk of driver damage and noisy ADC readings during Test 3 and Test 2 operations.

Power system upgrade: The clearest route to resolving the mobility limitations is a higher capacity battery (recommended: 48 V, ≥ 8 Ah) with a proper BMS to supply the necessary peak currents without excessive voltage sag and to protect cells from over-discharge. This also increases runtime and reduces the frequency of driver protection trips under normal use.

Control improvements: While classical PID worked well in many cases, oscillations increased at higher duty cycles; future work should explore adaptive gain scheduling or auto-tuning routines so PID gains adapt to rider mass and operating point. Also consider adding rotor-position encoders or sensor fusion with IMU to improve feedback quality for tighter control at higher speeds or heavier loads.

Robust driver selection: For a production design, use a motor controller with native external PWM/PWM-to-torque support and documented control interface to avoid fragile PCB modifications. That reduces the single-point failure risk we observed and simplifies integration of higher-level control algorithms.

Measurement and validation: Replace the simple voltage divider with a dedicated battery-monitoring module (or an ADC front end with RC filtering and buffering) to reduce noise and get reliable State-of-Charge estimates; use current measurement (shunt + amplifier) to log actual motor currents and correlate trips to peak currents.

4.4.7 SUMMARY AND FINAL REMARKS

The project validated the literature-based hypothesis that a low-cost PT can be implemented with BLDC hub motors, a microcontroller implementing PID, and a compact motor driver. Our implementation demonstrated correct closed-loop control behaviour for the target motor/wheel at moderate loads and speeds; important failure modes (battery sag, driver protection, fragile PCB mods, sensor noise) were uncovered in practical testing and quantified. The experimental evidence aligns with literature expectations in steady-state cruise conditions for light riders but diverged under heavy transient loads because of limitations in the power supply and conservative protection of low-cost motor drivers. The work therefore fulfilled its aim as a prototype demonstration while also delivering a clear roadmap (battery upgrade, driver robustness, adaptive control, improved sensing) for making the system practical and reliable in real-world deployment.

CHAPTER 5: CONCLUSION AND RECOMMENDATIONS

5.0 CONCLUSION

The successful completion of this project demonstrates the feasibility of developing a low-cost, locally sourced personal transporter (PT) using widely available components and fundamental engineering principles. The central aim was to design and implement a safe, reliable, and functional prototype capable of carrying an average rider while ensuring simplicity of operation and maintainability. Through an iterative design process that involved conceptualization, prototyping, testing, and evaluation, the project achieved its core objectives while also highlighting important insights for future improvements.

From a mechanical standpoint, the design calculations proved that the selected 6.5-inch hub motors were adequate to propel a rider weighing up to 70 kg along with the transporter's own mass of 7 kg. Comparative analysis with larger 8-inch and 10-inch hub motors demonstrated that although they provide higher torque and load capacities, their increased cost and sourcing difficulties made them less viable for this application. By defending the selection of the 6.5-inch motors with rigorous calculations and real-world mobility tests, the project not only validated the design choice but also emphasized the importance of balancing performance, cost, and availability in practical engineering design.

On the electrical side, careful consideration was given to component selection and integration. A 36 V lithium battery was chosen as the main power source due to its availability and cost-effectiveness, even though its 2.2 Ah capacity resulted in limited runtime. The ZS-X11H motor driver, rated at 400 W, was employed to control the brushless hub motors, offering a compact yet powerful solution with built-in protections. The Arduino Uno was used as the central controller, enabling implementation of PID control, communication with sensors, and feedback display on the LCD. Supporting components such as potentiometers, fuses, switches, and appropriate wiring gauges were integrated to ensure both safety and usability. These choices illustrate the project's focus on practicality, safety, and affordability while ensuring that the system could be easily maintained or modified in future iterations.

Testing and implementation were critical stages of the project, as they validated the performance of both the mechanical and electronic systems. The PID bench tests conducted in alignment with reference [35] allowed us to characterize motor feedback, define the operational limits, and test closed-loop speed control. Although challenges such as noise in the voltage divider readings and the risk of motor driver failure during modifications were encountered, these experiences provided valuable lessons for safer and more reliable design practices in the future. Additionally, the mobility tests revealed that while the PT operated effectively for riders up to 60 kg, performance declined for heavier riders, pointing to the need for a higher-capacity power supply or stronger motor drivers in future designs.

The system prototyping process also played a significant role in validating design concepts. User feedback confirmed that intuitive steering, centralized control components, and battery level monitoring added significant value to the design. At the same time, limitations such as the lack of advanced user interactions and non-adjustable handlebar height highlighted opportunities for refinement in future iterations. These insights reinforced the importance of involving users in the design cycle and provided a pathway for gradual improvement of the PT.

In conclusion, the project successfully delivered a working personal transporter prototype that meets its intended purpose within the defined constraints. It showcased the integration of mechanical design, electrical systems, and control strategies into a cohesive product, while also balancing cost and resource limitations. The work further emphasized the value of iterative design, user feedback, and practical testing in engineering innovation. Moving forward, future research and development could focus on improving system robustness by incorporating higher-capacity batteries, enhancing driver protection, refining PID tuning methods, and offering more personalized user interactions such as adjustable handlebar heights or advanced digital interfaces. By addressing these areas, the PT could evolve into a commercially viable product capable of serving a wider range of users safely and reliably.

Ultimately, this project not only achieved its technical objectives but also provided the research team with critical hands-on experience in problem-solving, interdisciplinary integration, and iterative design. These lessons form a strong foundation for future projects in personal mobility systems and demonstrate how practical engineering solutions can be developed even under constrained conditions.

5. 1 RECOMMENDATIONS

Based on the methodology adopted and the results obtained in this study, several recommendations are made to enhance the performance, safety, and usability of the three-wheeled personal transporter. These recommendations span improvements in control strategy, sensing, mechanical design, power management, and future research directions.

5.1.1 REFINEMENTS OF THE CONTROL SYSTEM

The application of the PID controller demonstrated clear improvements in system stability, disturbance rejection, and steady-state accuracy compared to open-loop and PI control. However, results also showed that oscillations began to appear at higher duty cycles (PWM values above 16). To address this limitation, it is recommended that adaptive or gain-scheduled PID controllers be investigated. These advanced strategies would allow the controller parameters to adjust dynamically depending on rider weight, terrain slope, or speed demands. This would ensure consistent performance across a wider range of operating

conditions, particularly when the transporter is subjected to sudden load variations such as riding uphill or carrying heavier users.

5.1.2 INTEGRATION OF HIGHER PRECISION SENSORS

Hall effect sensors built into the hub motors provided reliable speed feedback and were adequate for basic control. Nevertheless, their resolution is limited to one pulse per pole-pair transition, which can reduce precision at high speeds. Rotary encoders are recommended for applications requiring finer speed measurement and improved repeatability, especially if future designs target higher speeds or tighter control accuracy. Additionally, the integration of Inertial Measurement Units (IMUs) such as the MPU6050 would provide data on slope, tilt, and rider dynamics. This would not only enhance control but also enable advanced safety features like automatic slope compensation or warning alerts for unsafe rider posture.

5.1.3 IMPROVED POWER MANAGEMENT

The current voltage divider setup allowed basic monitoring of the 36 V battery pack by scaling down the voltage for the Arduino. While effective, this method is limited in accuracy and does not provide detailed information on battery health, capacity, or protection against over-discharge. It is strongly recommended that future iterations incorporate a Battery Management System (BMS). A BMS would offer precise monitoring of state-of-charge, temperature protection, and automatic cutoffs to prolong battery life and improve rider safety. This is particularly important given that battery failure or instability can lead to unsafe operation in transport systems.

5.1.4 MECHANICAL DESIGN OPTIMIZATION

The improvised frame, constructed using hoverboard hub motors and a welded bicycle wheel, achieved the desired functionality of a stable three-wheeled transporter. However, welding points and the overall frame design could be further optimized for long-term durability, especially under repeated mechanical stress. Use of lightweight but strong materials, such as aluminum alloys or reinforced composites, would reduce the overall weight of the transporter while maintaining strength. Improved ergonomics in handle placement and folding mechanisms are also recommended to enhance user comfort and portability, making the transporter more practical for everyday use.

5.1.5 SOFTWARE AND MICROCONTROLLER ENHANCEMENTS

The Arduino Uno successfully implemented the discretized PID controller, but its limited processing power and lack of connectivity restrict future scalability. Transitioning to a more

advanced microcontroller such as the ESP32 would provide greater processing capability, built-in wireless connectivity, and memory for logging operational data. This would enable additional features such as Bluetooth-based mobile applications for monitoring speed, battery percentage, and ride statistics. Real-time data logging could also support predictive maintenance and further research into rider behavior and system reliability.

5.1.6 EXTENDED TESTING UNDER REAL-WORLD CONDITIONS

The evaluation in this study was carried out in controlled environments. While sufficient for proof of concept, real-world applications introduce variables such as uneven terrain, weather conditions, long-term battery degradation, and varying rider weights. Extended field testing under diverse conditions is therefore recommended. Such tests would provide insight into robustness, rider comfort, and safety across realistic scenarios. This would also allow identification of potential design weaknesses that may not be apparent in laboratory-based experiments.

5.1.7 FUTURE RESEARCH DIRECTIONS

Finally, to further advance the project, future work could explore the integration of renewable energy charging options (e.g., solar-assisted charging), implementation of regenerative braking for efficiency, and the use of advanced control algorithms such as Model Predictive Control (MPC) or Fuzzy Logic. These approaches could improve both performance and sustainability, aligning the transporter with modern trends in green personal mobility solutions.

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APPENDICES

A. GANTT CHART

Tasks	8 Jan	24 Jan	21 Feb	18 Mar	5 Apr	13 May	10 Jun	14 Jun	7 Aug
Project title research									
Literature Review									
Design									
Collection of recyclable parts									
Simulations									
Buying of Components									
Fabrication									
Testing									
Assembling									
Final Product presented									

B. BUDGET

No.	NAME	QUANTINTY	COST	SOURCE
1.	Arduino and cable	1	K435	Locally
2.	2 ND hand small bike	1	K200	Locally
3.	Square tubes for frame	4 meters	K100	Locally
4.	Transport and miscellaneous	1	k1000	Locally
5.	I2C LCD	1	K295	Locally
6.	2 potentiometers 10k	1	K50	Locally

7.	830p breadboard	1	K125	Locally
8.	Motor drivers	4	K1300	China(eBay)
9.	Rocker Switch	1	K15	Locally
10.	Frame and headset		K2000	Locally
11.	Blade type fuse and holder		K25	Locally
	Total		K5545	

C. MATLAB CODE GENERATING GRAPH COMPARING PID, PI AND OPEN LOOP.

% Comparison of Open Loop, PI, and PID Control

clc; clear; close all;

% Define the plant (transfer function of motor system, for example)

% Assume a simple first-order system: $G(s) = 1 / (s+1)$

$s = tf('s');$

$G = 1 / (s + 1);$

% Controllers

$K_p = 2;$ % Proportional gain

$K_i = 1;$ % Integral gain

$K_d = 0.5;$ % Derivative gain

% Open loop (just plant)

$sys_open = G;$

```
% PI controller:  $C(s) = K_p + K_i/s$ 
```

```
C_PI = Kp + Ki/s;
```

```
sys_PI = feedback(C_PI*G, 1);
```

```
% PID controller:  $C(s) = K_p + K_i/s + K_d*s$ 
```

```
C_PID = Kp + Ki/s + Kd*s;
```

```
sys_PID = feedback(C_PID*G, 1);
```

```
% Step responses
```

```
figure;
```

```
step(sys_open, 'r', sys_PI, 'b', sys_PID, 'g', 10);
```

```
legend('Open Loop', 'PI Control', 'PID Control');
```

```
title('Step Response Comparison: Open Loop vs PI vs PID');
```

```
xlabel('Time (s)');
```

```
ylabel('Response');
```

```
grid on;
```

D. TABLE SHOWING WIRE GAUGE DIAMETER AND APPLICATION

Wire Gauge	AWG Diameter (mm)	SWG Diameter (mm)	Typical Applications
10	2.5883	3.251	Heavy-duty electrical wiring, industrial machinery
11	2.3038	2.946	Automotive battery cables, heavy appliances

12	2.0523	2.642	Residential wiring for high-current applications
13	1.8288	2.337	Medium-duty wiring, loudspeakers, small appliances
14	1.6281	2.032	Household electrical wiring, lighting circuits
15	1.4503	1.829	Low-current circuits, intercom systems
16	1.2903	1.626	Extension cords, small motors, low-power devices
17	1.1506	1.422	Electronic devices, internal wiring
18	1.0236	1.219	Small household devices, thermostats, doorbells
19	0.9119	1.016	Telephone lines, hobby electronics
20	0.8128	0.914	Signal cables, low-power circuits, data transmission
21	0.7239	0.813	Low-voltage lighting, control wiring
22	0.6426	0.711	Audio cables, microcontrollers, general electronics
23	0.574	0.61	Sensor wiring, precision equipment

24	0.5105	0.599	Computer peripherals, fine wiring for electronics
25	0.4547	0.508	Data cables, LED lighting strips
26	0.4039	0.457	Instrumentation wiring, medical devices
27	0.3607	0.417	Fine electronics, headphones
28	0.32	0.376	Ultra-fine wiring, precision instruments
29	0.287	0.345	Small transformers, delicate equipment
30	0.254	0.315	Miniature electronics, toys, ultra-thin applications

E. STEP RESPONSE FOR UNCOMPENSATED

```

1      %UNCOMPENSATED STEP RESPONSE CODE
2
3      clc
4      clear
5
6      %UNCOMPENSATED PLANT
7      s = tf('s');
8      numg = 0.04125;          % Numerator of G(s) for the plant.
9      deng = [0.0002 0.14544 0.272]; % Denominator of G(s) for the plant.
10     G = tf(numg, deng)      % Transfer Function
11
12     %UNCOMPENSATED STEP RESPONSE
13     step(G);
14     title('UnCompensated System Step Response')
15     grid on

```


F. POLE-ZERO PLOT CODE FOR UNCOMPENSATED STEP RESPONSE

```

1      % Define numerator and denominator of the transfer function
2      num = 0.869;
3      den = [0.000071 0.03551 0.077127];
4
5      % Create transfer function
6      G = tf(num, den);
7
8      % Plot pole-zero map
9      figure;
10     pzmap(G);
11     grid on;
12     % Define numerator and denominator of the transfer function
13     num = 0.869;
14     den = [0.000071 0.03551 0.077127];
15
16     % Create transfer function
17     G = tf(num, den);
18
19     % Bode plot
20     figure;
21     bode(G);
22     grid on;
23     title('Bode Plot of the Uncompensated Motor Plant');
24
25     % Nyquist plot
26     figure;
27     nyquist(G);
28     grid on;
29     title('Nyquist Plot of the Uncompensated Motor Plant');

```

H. COMPENSATED STEP RESPONSE

```

clc;
clear;

%% Define s
s = tf('s');

%% UNCOMPENSATED PLANT (Original transporter plant)
numG = [0.869];
denG = [0.000071 0.03551 0.077127];
G = tf(numG, denG);

%% COMPENSATED PLANT (Initial Compensation)
% Here we can keep the structure, assuming same as before (adjusted for transporter)
% For simplicity, let's keep numG2/denG2 same scale as original example
numG2 = [0.869 0.5 0.1]; % example initial compensation numerator
denG2 = [0.000071 0.03551 0.077127 0]; % adds an integrator for compensation
G2 = tf(numG2, denG2);

```

```

%% Step Response Comparison
t = 0:0.001:2; % smaller time vector due to fast response

[y1, t1] = step(G, t);
[y2, t2] = step(G_F1, t);
[y3, t3] = step(G_F2, t);

figure;
plot(t1, y1, 'b', 'LineWidth', 1.5); hold on; % Uncompensated
plot(t2, y2, 'r--', 'LineWidth', 1.5); % Initial Compensated
plot(t3, y3, 'g-', 'LineWidth', 1.5); % Fine-Tuned Compensated

grid on;
xlabel('Time (s)');
ylabel('Angular Velocity (rad/s)');
title('Step Response Comparison: Uncompensated vs Compensated vs Fine-Tuned');
legend('Uncompensated', 'Compensated', 'Fine-Tuned Compensated');
hold off;

```

I. NYQUIST, BODE PLOT AND ROOT LOCUS FOR COMPENSATED SYSTEM

```

clc;
clear;

%% Define s
s = tf('s');

%% =====
%% UNCOMPENSATED PLANT (Original transporter plant)
numG = [0.069];
denG = [0.000071 0.03551 0.077127];
G = tf(numG, denG);

%% =====
%% COMPENSATED PLANT (Initial compensation, illustrative)
numG2 = [0.069 0.5 0.1]; % initial compensation numerator
denG2 = [0.000071 0.03551 0.077127 0]; % adds an integrator
G2 = tf(numG2, denG2);

G_F1 = feedback(G2, 1); % Unity feedback

%% =====
%% FINE-TUNED COMPENSATED SYSTEM (PID)
Kp = 980.4369;
Ki = 489.96845;

```

```

G_F2 = feedback(C*G, 1); % Fine-tuned PID with unity feedback

%% =====
%% Step Response Comparison
t = 0:0.001:2; % time vector

[y1, t1] = step(G, t); % Uncompensated
[y2, t2] = step(G_F1, t); % Initial Compensated
[y3, t3] = step(G_F2, t); % Fine-Tuned Compensated

figure;
plot(t1, y1, 'b', 'LineWidth', 1.5); hold on;
plot(t2, y2, 'r--', 'LineWidth', 1.5);
plot(t3, y3, 'g-.', 'LineWidth', 1.5);
grid on;
xlabel('Time (s)');
ylabel('Angular Velocity (rad/s)');
title('Step Response Comparison: Uncompensated vs Compensated vs Fine-Tuned');
legend('Uncompensated', 'Compensated', 'Fine-Tuned Compensated');
hold off;

%% =====
%% ROOT LOCUS
figure;
rlocus(G);
title('Root Locus - Uncompensated Plant');
grid on;

%% =====
%% BODE PLOTS
figure;
bode(G);
title('Bode Plot - Uncompensated Plant');
grid on;

figure;
bode(C*G);
title('Bode Plot - Compensated Plant (PID)');
grid on;

%% =====
%% NYQUIST PLOTS
figure;
nyquist(G);
title('Nyquist Plot - Uncompensated Plant');
grid on;

figure;
nyquist(C*G);
title('Nyquist Plot - Compensated Plant (PID)');
grid on;

```

J. %EFFICACY CODE

```
clc;
clear;

%% Sampling time
T = 0.01; % 10 ms

%% Discrete PID transfer function using your coefficients
numgz = [1180.84 -2356.88 1178.4]; % numerator of X(z)/E(z)
dengz = [1 -1 0]; % denominator of X(z)/E(z)
Gz = tf(numgz, dengz, T); % discrete-time transfer function

%% Time vector
time = 0:T:5;

%% Step input simulation
step_input = ones(size(time));
[Y_step, T_step] = lsim(Gz, step_input, time);

figure;
plot(T_step, Y_step, 'b-', 'LineWidth', 1.5);
hold on;
plot(T_step, step_input, 'r--', 'LineWidth', 1.5);
title('System Response to Rectangular Trajectory (Step Input)');
xlabel('Time (s)');
ylabel('Output');
legend('System output', 'Reference Signal');
grid on;

%% Sinusoidal input simulation
frequency = 1; % 1 Hz
circular_input = sin(2*pi*frequency*time);
[Y_circular, T_circular] = lsim(Gz, circular_input, time);

figure;
plot(T_circular, Y_circular, 'b-', 'LineWidth', 1.5);
hold on;
plot(T_circular, circular_input, 'r--', 'LineWidth', 1.5);
title('System Response to Circular Trajectory (Sinusoidal Input)');
xlabel('Time (s)');
ylabel('Output');
legend('System output', 'Reference Signal');
grid on;
```

K. MIN AND MAX FEEDBACK (TEST 1) ARDUINO CODE

```
// MinMax - 10/02/2024 - reference version

int HALL_PIN = 2;           // interrupt feedback pin
int PWM_PIN_OUT = 3;        // 490Hz PWM Output
int POT_PIN_IN = A7;        // Analog potentiometer for speed

volatile float feedback=0.00, tic, tac;           // interrupt function variables
float pot=0, target, trgt_min=0, trgt_max=255, fb_min=0, fb_max=0; // target_PWM value
- min and max values between target_PWM and feedback

void setup() {
  Serial.begin(115200);           // serial monitor
  pinMode(PWM_PIN_OUT, OUTPUT);   // 490Hz PWM Output
  pinMode(POT_PIN_IN, INPUT);     // potentiometer
  attachInterrupt(digitalPinToInterrupt(HALL_PIN), intrupt, RISING); // Attach Interrupt
  for motor Hall sensors
}

void loop() {
  pot = analogRead(POT_PIN_IN);   // throttle value, determined by potentiometer
  target = map(pot,0,1023,trgt_min,trgt_max); // convert throttle in PWM value
```

```

analogWrite(PWM_PIN_OUT, target);          // output PWM target

Serial.print("\n target: ");Serial.print(target,0); Serial.print(" feedback:
");Serial.print(feedback,0); delay(200);
}

void intrupt(){
    tic = millis();                        // time

    feedback = tic - tac;                  // time between 2 tic

    tac = tic;                            // save previous time
}

```

L. PID CONTROLLER CODE (TEST 2)

```

➤ // HB_Motor_PID - with PWM and ZS-X11H controller - 10/02/2024
➤ int pb = A0;                          // Start/Stop pgm button
➤ int DIR_PIN_BT = A2;                  // Direction button
➤ int DIR_PIN_OUT = 10;                 // Direction Output
➤ int HALL_PIN = 2;                     // interrupt feedback pin
➤ int PWM_PIN_OUT = 3;                 // 490Hz PWM Output
➤
➤ volatile float tic, tac, feedback=0.00; // interrupt variables
➤ bool dir=0;                          // direction, 0=clockwise,
1=counterclockwise
➤ float now, prvTime, dt;               // time variables
➤ float P=0.00, I=0.00, D=0.00, error=0.00, errDiff=0.00, prevErr=0.00,
maxSum=50, errSum=0.00, pid=0.00; // PID variables
➤
➤ float kp=0.15, ki=0.7, kd=0.001, target=15, trgt_min=12, trgt_max=25,
fb_min=104, fb_max=46; // variables to be modified
➤
➤ void setup() {
➤     Serial.begin(115200);
➤     pinMode(pb, INPUT_PULLUP);
// Start pgm button
➤     pinMode(DIR_PIN_BT, INPUT_PULLUP);
// Direction button
➤     pinMode(DIR_PIN_OUT,
OUTPUT); // Direction Output

```

```

➤   pinMode(PWM_PIN_OUT,
OUTPUT);                                     // 490Hz PWM Output
➤   attachInterrupt(digitalPinToInterrupt(HALL_PIN), intrupt,
RISING);                                     // Attach Interrupt for motor Hall sensors
➤   Serial.print("\n\t To start pgm, click on the Start button");
while(digitalRead(pb)); delay(400); Serial.print("\n\t Started"); // start
button
➤   }
➤
➤   void loop() {
➤       now = millis(); dt = (now - prvTime) /1000.00; prvTime =
now;                                         // time between two loops
➤       if( ! digitalRead(DIR_PIN_BT)){ dir = dir ^ 1; delay(300);
analogWrite(PWM_PIN_OUT,0); digitalWrite(DIR_PIN_OUT, dir); } // change
direction when button pushed
➤       pid =
PID();                                     // PID
calculation
➤       analogWrite(PWM_PIN_OUT, round(pid = constrain(pid,trgt_min,trgt_max))
); // output PWM PID - constrain speed for security
➤       Plotter();
// values for the plotter
➤       //Trace();
// print variables
➤   }
➤
➤   void intrupt(){
➤       tic = millis();
➤       feedback = tic - tac;  tac = tic;                                         //
time between 2 Hall sensor detections
➤       feedback = map( feedback, fb_min, fb_max, trgt_min, trgt_max ); //
convert feedback milliseconds to PWM value
➤   }
➤
➤   float PID(){
➤       noInterrupts(); error = target - feedback; interrupts();
➤       P = kp * error;
➤       I = ki * (errSum = errSum + (error * dt)); errSum = constrain( errSum, -
maxSum, maxSum );
➤       D = kd * (error - prevErr) / dt; prevErr = error;
➤       return P + I + D;
➤   }
➤
➤   void Plotter(){

```

```

➤ Serial.print(0);          Serial.print(" ");
  // to limit plotter scale
➤ Serial.print(feedback,3);  Serial.print(" ");
➤ Serial.print(pid,3);       Serial.print(" ");
➤ Serial.println(trgt_max,0);
  // to limit plotter scale
➤ }
➤
➤ void Trace(){
➤   Serial.print(String() + "\n"
➤               + " target: " + target
➤               + " feedback: " + String(feedback,3)
➤               + " pid: " + String(pid,3)
➤               + " error: " + String(error,3)
➤               + " prevErr: " + String(prevErr,3)
➤               + " errSum: " + String(errSum,3)
➤               + " P: " + String(P,3)
➤               + " I: " + String(I,3)
➤               + " D: " + String(D,3)
➤               + " dir: " + dir
➤               + " dt: " + String(dt,3)
➤               );
➤ }
➤

```

M. LCD/ BATTERY MONITORING ARDUINO CODE (TEST 3)

```

N. #include <Wire.h>
O. #include <LiquidCrystal_I2C.h>
P.
Q. // LCD address 0x27 or 0x3F
R. LiquidCrystal_I2C lcd(0x27, 16, 2);
S.
T. const int voltagePin = A0;      // Analog input pin
U. const float ratio = 9.2;        // Voltage divider ratio
V.
W. void setup() {
X.   Serial.begin(9600);
Y.   lcd.init();
Z.   lcd.backlight();
AA.  lcd.clear();
BB. }
CC.
DD. void loop() {
EE.  float BatteryV = readVoltage(voltagePin, ratio);

```



```

FF.
GG. // Print to Serial
HH. Serial.print("Battery Voltage: ");
II. Serial.print(BatteryV, 2);
JJ. Serial.println(" V");
KK.
LL. // Print to LCD
MM. lcd.setCursor(0, 0);
NN. lcd.print("Battery:      ");
OO. lcd.setCursor(0, 1);
PP. lcd.print(BatteryV, 2);
QQ. lcd.print(" V          "); // Clear leftover characters
RR.
SS. delay(1000);
TT.}
UU.
VV.// -----
WW.// Function to read voltage
XX.// -----
YY.float readVoltage(int pin, float dividerRatio) {
ZZ.  const float Vref = 5.0;           // Arduino Uno analog reference
    voltage
AAA.      const float ADC_Resolution = 1023.0;
BBB.
CCC.      int raw = analogRead(pin);
DDD.      float vOut = (raw * Vref) / ADC_Resolution;
EEE.      float vIn = vOut * dividerRatio;
FFF.      return vIn;
GGG.  }

```